

Powered Lift Wing Design for STOL Applications

End Evaluation Report

InterIIT Tech Meet 14.0 - High Prep LAT Aerospace

TEAM 76

A multi-fidelity aerodynamic-propulsive design framework was developed to maximize lift through coordinated use of high-lift devices and propulsion-induced flow control, enabling a STOL-capable configuration that leverages blown-wing effects, refined flap geometry, and propeller-wing interaction analysis.



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Abbreviations

Abbreviation	Full Form
STOL	Short T ake- O ff and L anding
USB	Upper S urface B low (wing)
EBF	Externally B low F laps
IDBF	Internally D ucted B low F laps
CFD	Computational F luid D ynamics
AoA	Angle of A ttack
AR	Aspect R atio
SA	Spalart-Allmaras T urbulence M odel
LE	Leading E dge
TE	Trailing E dge
Fluent	A NSYS F luent (Flow Solver)
ADT	Actuator D isc T heory
UDF	User D efined F unction
csv	Comma Separated V alue

Symbols

Symbol	Description
a	Axial induction factor
a'	Tangential induction factor
a_0	Two-dimensional lift-curve slope
a_{3D}	Three-dimensional lift-curve slope
A_0, B_0	Fourier coefficients (Jet Flap Theory)
A_d	Actuator disk area
A_w	Wake cross-sectional area
A_∞	Freestream cross-sectional area
B	Number of blades
c	Chord length
c_{tip}, c_{root}	Tip chord and Root chord
C_D	Coefficient of Drag
$C_{D,i}$	Induced Drag Coefficient
C_f	Skin friction coefficient
C_J	Jet momentum coefficient
C_L	Coefficient of Lift
$C_{L,max}$	Maximum Coefficient of Lift
C_l, C_d	Sectional lift and drag coefficients
C_m	Pitching moment coefficient
C_n, C_t	Normal and Tangential force coefficients
C_p	Pressure coefficient
$C_{Z,raw}$	Total vertical force coefficient (Slipstream)
e	Oswald's Efficiency Factor
F	Tip loss correction factor
f	Body forces per unit volume
f_s, f_{w0}	Viscous loss factors
J	Momentum flux per unit span
$\mathcal{L}$	Loss function (Neural Network)
M	Mach number
p	Pressure field
Q	Torque
Re	Reynolds number
T	Thrust
u, v, w	Velocity vector components
U_∞, V_∞	Free-stream velocity
x_{ac}	Aerodynamic center location
x_{ref}	Reference location along chord
y^+	Non-dimensional wall distance
α	Angle of Attack
$\epsilon_s, \epsilon_{w0}$	Flow deflection angles
θ	Twist Angle or Flap orientation
λ	Taper Ratio
μ	Dynamic viscosity
ν	Kinematic viscosity
ρ	Air density
τ	Jet deflection angle

Executive Summary

Overview and Design Approach

The LAT Aerospace STOL project was initiated to address the growing demand for short take-off and landing capabilities in both civil and defense sectors. As urbanization increases and the need for disaster relief and surveillance operations in confined environments grows, the ability to operate from short, unprepared runways has become a critical performance metric. This project specifically targets the development of a high-lift wing configuration that seamlessly integrates conventional high-lift devices with advanced propulsion-assisted lift mechanisms. The core design philosophy centers on a coupled aerodynamic-propulsive approach, moving beyond purely geometric lift enhancement to actively energize the flow field. This methodology aims to reconcile the often conflicting requirements of maximizing lift coefficients at low speeds while maintaining structural efficiency and aerodynamic stability during cruise. To achieve these ambitious performance goals, the team adopted a rigorous, multi-fidelity design methodology. The process began with foundational theoretical modeling, utilizing Blade Element Theory (BET) to estimate propulsion performance and Jet-Flap theory to predict wing behavior under blown conditions. These lower-order models provided rapid initial sizing and were subsequently validated and refined through extensive Computational Fluid Dynamics (CFD) simulations. The team utilized ANSYS Fluent with the Spalart-Allmaras turbulence model to capture complex flow phenomena, such as boundary layer separation and wake interactions. Furthermore, the project explored cutting-edge optimization techniques by implementing Physics-Informed Neural Networks (PINNs). This data-driven approach allowed for rapid exploration of the design space, specifically for multi-element airfoil configurations, creating a robust mathematical model to guide the final geometric refinements.

Airfoil Selection and High-Lift Development

The foundation of the aerodynamic design was the selection of the Selig S1223 airfoil. After a comprehensive comparative analysis against alternative profiles like the Selig S1221 and E426, the S1223 was chosen for its superior suction peak characteristics and inherently high maximum lift coefficient ($C_{L,max}$), which approached 2.0 in baseline single-element simulations. Its geometric camber and thickness distribution made it particularly well-suited for low Reynolds number operations, a critical regime for STOL aircraft during take-off and landing. This baseline performance provided a strong platform for the integration of high-lift devices.

Following the airfoil selection, the team conducted an exhaustive parametric study of flap configurations to maximize lift generation. This investigation examined the effects of flap chord length, deflection angle, and the gap/overlap positioning relative to the main element. The simulations revealed that a flap with a chord length of 40% of the main airfoil produced the highest lift coefficients, achieving values near 2.49. However, this came with a significant drag penalty. Conversely, a 20% chord flap offered a more balanced lift-to-drag profile. The study also highlighted the sensitivity of the flow to the gap geometry; optimal lift was consistently found in regions where the flap was positioned moderately downstream and slightly above the main trailing edge, balancing high-energy flow injection with wake management.

Propulsion System Integration and Propulsive Lift

A central innovation of this design is the utilization of distributed propulsion to actively augment lift. The team selected an open-propeller architecture over Electric Ducted Fans (EDFs) to leverage the beneficial effects of a broad, uniform slipstream over the wing surface. The propulsion units were modeled using Virtual Blade Models (VBM) within the CFD environment, allowing for a computationally efficient yet accurate representation of the propeller's momentum and swirl effects. This approach was critical in evaluating the complex interactions between the propeller wake and the wing boundary layer.

The investigation into propulsive lift mechanisms yielded significant insights regarding the placement of propulsion units. A series of vertical and longitudinal position studies determined that mounting the propellers above the wing chord line (positive vertical offset) significantly enhanced the suction on the upper surface, thereby increasing the lift coefficient. However, this configuration also introduced

challenges regarding flow stability and efficiency. While the elevated position maximized lift, it exacerbated vortex shedding at high angles of attack. Consequently, a compromise position was recommended approximately 10 cm upstream of the leading edge with a moderate vertical offset to balance the lift benefits of Upper Surface Blowing (USB) with the need for stable flow attachment and acceptable drag levels.

Furthermore, the integration of internally ducted blown flaps proved to be a decisive factor in achieving the project's high-lift targets. Initial designs utilizing a forward-located duct exit resulted in severe flow separation due to the adverse pressure gradients near the leading edge. By shifting the duct exit to an aft position (Case III), the team successfully managed the laminar-to-turbulent transition, maintaining boundary layer attachment over a much larger portion of the chord. This geometric optimization resulted in a non-linear improvement in aerodynamic performance, drastically increasing the lift capabilities of the system without requiring larger external control surfaces.

3D Wing Performance and Stability Analysis

Transitioning from 2D profiles to a full 3D wing configuration introduced new challenges related to induced drag and spanwise flow effects. The baseline 3D wing successfully met the static lift requirements, achieving a lift coefficient of 6.6. However, it initially fell short of the aerodynamic efficiency targets, with a lift-to-drag ratio (L/D) of only 4.55. This prompted a second design iteration that refined the high-lift system and propulsor shroud geometry. By adjusting the tertiary flap deflection angles and optimizing the propulsor intake, the team was able to boost the lift coefficient to 7.67 and improve the efficiency ratio to approximately 6.4.

Despite meeting the numerical targets, the high-lift configuration exhibited signs of aerodynamic instability. Detailed transient simulations revealed periodic vortex shedding from the trailing edges and flap gaps at maximum lift conditions. This unsteadiness indicates that while the wing generates exceptional static lift, the flow field is on the verge of separation, creating a narrow operational envelope. The parametric sensitivity analysis confirmed that the design is highly sensitive to propeller placement; minor deviations in vertical or longitudinal positioning could trigger immediate flow detachment or solution divergence. Therefore, while the performance targets have been technically achieved, future work must focus on stabilizing the wake structure to ensure safe and predictable handling characteristics during critical flight phases.

Performance Achievements

The executive summary concludes with a quantitative review of the project's achievements. The table below details the specific parameter values obtained through simulation and analysis, contrasting them against the initial design requirements set forth in the problem statement.

Parameter	Configuration / Context	Achieved Value	Status
Max Lift Coefficient ($C_{L,max}$)	Single Element Airfoil (S1223)	1.9998	Baseline Established
Max Lift Coefficient ($C_{L,max}$)	2D Airfoil with 40% Flap	2.4855	Optimized
Max Lift Coefficient ($C_{L,max}$)	Internally Ducted (Case III - Aft)	4.7718	High Performance
Lift-to-Drag Ratio (L/D)	Internally Ducted (Case III - Aft)	15.65	Efficient
Takeoff Lift Coefficient (C_L)	3D Wing (Baseline)	6.8195	Target Met
Takeoff Efficiency (L/D)	3D Wing (Baseline)	5.37	Target Met
Takeoff Lift Coefficient (C_L)	3D Wing (Refined High-Lift System)	7.67	Target Met*
Takeoff Efficiency (L/D)	3D Wing (Refined High-Lift System)	~ 6.39	Target Met*

**Note: The Refined High-Lift System achieved the efficiency target but exhibited transient flow instabilities (vortex shedding) that require further stabilization.*

Introduction

A Short Take-Off and Landing (STOL) aircraft is one capable of performing take-off and landing within very short distances (less than commercial aircraft but still more than VTOLs and drones). It combines the efficiency and the practicality of having a fixed wing and fixed rotor aircraft, while improving the take-off and landing performance to expand the flight envelope. For cities worldwide with capabilities that support only small areas for aircraft, STOLs are a promising alternative to having large jets. The development of High-Lift Devices (HLDs), as well as various propulsive lift devices, can be easily extended to build the next generation of highly efficient, customer-centric aircraft, with possible applications in defense as well. These aircraft are commonly equipped with multiple HLDs including flaps, slats, etc. They are aided by powered lift systems like upper surface blown wings, externally and internally blown flaps, supercirculation wings, among others. Working majorly on the Coanda effect, flaps and slats effectively manipulate airflow to improve the coefficient of lift, although taking a huge hit in terms of drag values.

The problem statement centered on developing a high-lift wing configuration capable of achieving enhanced lift performance while maintaining aerodynamic efficiency. Therefore, the primary performance goal involved maximizing the C_L and improving the C_L/C_D . The challenge was in achieving high lift without inducing excessive drag or flow separation, while ensuring compatibility with the single propulsive plant. This required a design approach that increased lift using aerodynamic methods alone, without depending on any additional thrust system. Based on this understanding, the team developed a structured approach that combines theoretical modeling, computational analysis, and design validation.

Design Methodology and Approach

The team initially focused on identifying suitable 2D airfoils with inherently high lift characteristics. This was done with the objective of achieving a high initial lift coefficient and satisfactory L/D ratios as these are significantly hit during propulsive lift. High lift devices such as flaps and slats were explored to have an increment and control over lift and lift-to-drag ratios. The designs and positions were decided using various strategies to achieve optimality. In parallel to HLDs, Internal Blown Flap strategy was employed to achieve high lift coefficient without the invocation of any other thrust plant. To validate the design and configuration, Jet flap theory was employed to validate the CFD simulations, ensuring consistency between analytical predictions and numerical outcomes.

Propellers were also being designed to achieve the required thrust values. Blade Element Theory (BET) was used to obtain geometry of the propellers with various input parameters such as free stream velocity and thrust. The BET calculations were initially verified and optimized using open source software such as QBlade. These results were then validated using CFD analysis using a custom User Defined Function (UDF) for Actuator Disk Theory (ADT), followed by the standard Virtual Blade Method (VBM).

After getting satisfactory results with the 2D airfoils, 3D wing designs were simulated. This was done to increase the efficiency of results, providing a larger data set to analyze. The HLD dataset was used to create a case-specific mathematical model in the form of a Physics Informed Neural Network (PINN). This was accompanied by theoretical mathematical models like Jet-Flap theory for wing performance and their comparison with CFD data. In addition, the aerodynamic center was also identified in order to aid stability analysis and control surface design of the wing.

Historical Context and Technological Evolution

What was once thought possible only through lighter-than-air systems such as balloons and blimps was revolutionized by the discovery of fixed-wing lift generation, an invisible upward force that redefined transportation and exploration. Over the decades, this pursuit of aerodynamic efficiency has driven a continuous evolution from propeller-driven aircraft to turboprops, turbojets, ramjets, and scramjets, each advancement representing a leap in propulsion capability. Today, the aerospace industry stands at the threshold of hybrid and distributed propulsion systems, concepts once deemed infeasible, now becoming central to the next generation of sustainable and high-performance aircraft.

From the fundamentals of fluid mechanics to advanced turbulence modeling and transient analysis, Computational Fluid Dynamics (CFD) has emerged as a powerful tool for aerodynamic exploration. When combined with empirical validation through wind tunnel experiments, aircraft systems can now be almost completely analyzed, optimized, and verified under simulated conditions that closely replicate reality. Validation of CFD simulations is a critical step to ensure that numerical predictions faithfully represent physical behavior. Typically, this process involves comparing simulated aerodynamic coefficients with results derived from analytical, empirical, or experimental models. Deviations between CFD and analytical predictions provide insight into the influence of viscous effects, flow separation, and compressibility—phenomena often neglected in theoretical formulations.

Software Declaration

- ANSYS 2024R1 license was available on an Enterprise level server license provided by the institution.
- MATLAB license is also provided by the institution.
- Autodesk FUSION 360 license is also provided by the institution.

Airfoil Selection

1.1 Objective and Selection Criteria

The aerodynamic performance of a STOL aircraft is strongly influenced by its airfoil characteristics, particularly in low Reynolds number regimes encountered during take-off and landing. As part of the project, the objective of this section is to identify an airfoil capable of producing a high maximum lift coefficient while maintaining acceptable aerodynamic efficiency and stable stall behavior. The aim at this stage is to establish a reliable baseline airfoil for subsequent high-lift device integration, propulsion interaction studies, and three-dimensional wing development.

The following parameters were used to evaluate candidate airfoils:

- Maximum lift coefficient, $C_{L,\max}$
- Angle of attack at $C_{L,\max}$
- Maximum lift-to-drag ratio, $(C_L/C_D)_{\max}$
- Lift coefficient at $(C_L/C_D)_{\max}$
- Performance consistency at low Reynolds numbers

1.2 Preliminary Shortlisting

An initial pool of approximately twenty airfoils was selected based on their historical usage in low-speed and STOL aircraft. Aerodynamic data from publicly available databases and previous studies was used for preliminary screening.

From this initial set, six airfoils (E426, E387, E420, S1221, S1223, M06-13-128) were shortlisted based on their attainable maximum lift coefficient. Airfoils that exhibited poor low Reynolds number performance or unfavorable stall behavior were eliminated. Further comparison using XFLR5 resulted in the selection of two final candidates:

- Selig S1221
- Selig S1223

It is important to note that the same analytical and numerical workflow was applied to both S1221 and S1223 airfoils. This included identical computational domains, meshing strategies, turbulence models, boundary conditions, and post-processing techniques. To maintain clarity and conciseness in this end-evaluation report, only representative CFD results for S1223 are presented in detail, while conclusions are drawn based on a direct comparison between both airfoils.

1.3 Comparison of Final Airfoils

The key aerodynamic and geometric characteristics of the final airfoils are summarized in Table [1.1](#).

Parameter	Selig S1221	Selig S1223
Maximum thickness (%)	12.1 at 21.8% chord	12.1 at 19.8% chord
Maximum camber (%)	4.9 at 49.1% chord	8.1 at 49% chord
Theoretical $C_{L_{max}}$	2.1 at $\alpha = 15^\circ$	2.2 at $\alpha = 13^\circ$
Maximum C_L/C_D	138 at $\alpha = 8^\circ$	120 at $\alpha = 5^\circ$
Type / Usage	Low Reynolds number applications	High-lift, low Reynolds number airfoil

TABLE 1.1: Comparison of Selig S1221 and S1223 Airfoil Characteristics

While both airfoils demonstrated high lift potential, the S1223 airfoil exhibited increased camber and stronger suction characteristics, making it more suitable for integration with high-lift devices and propulsive lift mechanisms.

1.4 CFD Methodology

1.4.1 Computational Domain

The computational domain was a **2D surface** with the airfoils of chord length, $L_c = 0.1\text{m}$ placed at the center. Clearances of $10L_c$ were set upstream and downstream. The boundary above and below the airfoil was placed $6L_c$ away on both sides. Named selections were created for the inlet, outlet, walls and airfoil edges for future reference.

1.4.2 Grid Independence

The mesh size was determined through a grid independence test. Three values were selected two close to each other and a third placed farther apart to evaluate the mesh quality across a wide range of element sizes and element counts. Mesh sizes of 4 mm, 5 mm, and 10 mm were chosen. **The 4 mm mesh represented the finest value due to its proximity to the last inflation layer.**

Mesh Size (mm)	Angle of Attack (α)	C_L	C_D
4	13.25°	1.9998	0.067369
5	13.25°	2.0054	0.067503
10	13.25°	2.0150	0.067468

TABLE 1.2: Airfoil S1223: Comparison of C_L and C_D for Various Mesh Sizes at Angle of Attack 13.25°

1.4.3 Turbulence Modelling

In this study, turbulence effects were modeled using **Reynolds-Averaged Navier–Stokes (RANS)** formulations, which provide a practical balance between accuracy and computational cost for engineering-scale aerodynamic simulations. The **Spalart–Allmaras** model was used for the simulations. The Spalart–Allmaras model is based on turbulence which involves eddy flows that include

mixed translational and rotational motion in a fluid. Some other RANS options were the standard $k-\varepsilon$ and SST $k-\omega$ turbulence models. But finally Spalart–Allmaras model was chosen due to it's application in external aerodynamic simulations.

Parameter	Specification 1	Specification 2
Inlet	Velocity Inlet	20m/s
Outlet	Pressure Outlet	0Pa Gauge Pressure
Turbulence Model	Spalart–Allmaras	Turbulence Viscosity Ratio 10
Reference Area	0.1 m ²	
Reference Length	0.1 m	
Report Definitions	Coefficient of Lift (C_L)	Coefficient of Drag (C_D)
Initialization Method	Hybrid Initialization	

TABLE 1.3: CFD Simulation Setup Parameters

1.4.4 Results

CFD simulations were carried out for both the S1221 and S1223 airfoils across a range of angles of attack and mesh resolutions. As the observed convergence behavior, mesh sensitivity trends, and qualitative flow features were similar for both airfoils, only the S1223 results are shown here as representative cases in order to avoid redundancy in the report.

Angle of Attack (α)	C_L	C_D
12.75°	1.8941	0.067071
13.25°	1.9998	0.067369
13.75°	1.8423	0.085883

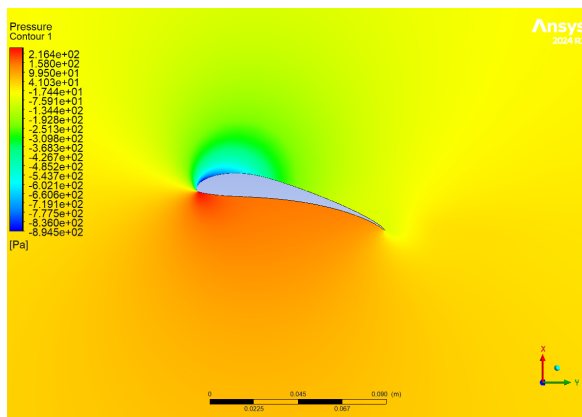
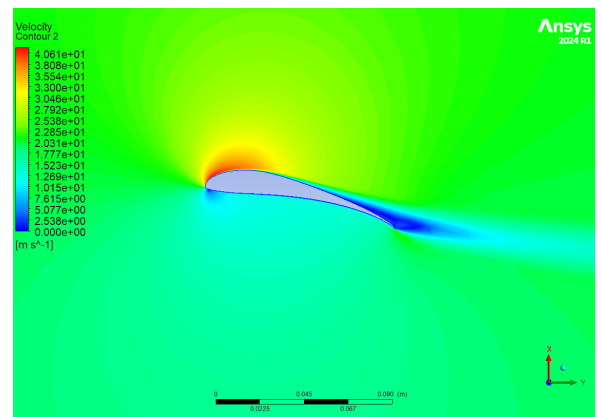
TABLE 1.4: C_L and C_D values at various angles of attack for 4 mm mesh with airfoil S1223(A) Pressure contours at $\alpha = 13.25^\circ$ (4 mm mesh)(B) Velocity contours at $\alpha = 13.25^\circ$ (4 mm mesh)

FIGURE 1.1: Representative pressure and velocity contours for the S1223 airfoil.

Quantitatively, the S1223 airfoil achieved a maximum lift coefficient of approximately $C_L=1.9998$ at an angle of attack close to 13.25° for the grid-independent mesh. The corresponding drag coefficient at this condition was approximately $C_D=0.067369$. Beyond this operating point, further increases in angle of attack resulted in a rapid rise in drag accompanied by a gradual reduction in lift, indicating the onset of stall behavior. These trends were consistently observed across all grid-independent simulations.

Overall, the S1223 airfoil demonstrated superior lift capability and sustained favorable flow attachment closer to stall conditions when compared to the S1221 airfoil. The stronger suction peak near the leading edge and the associated pressure recovery characteristics make it particularly suitable for integration with high-lift devices and propulsive lift mechanisms.

1.5 Re-evaluation of Airfoil Selection Criteria

Following the initial selection of the S1223 airfoil, a critical design review was conducted regarding the impact of power-augmented lift on aerodynamic efficiency. Feedback indicated that the initial selection criteria relied on conventional wing design approaches which did not adequately account for the aerodynamic coupling between the propulsion system and the wing. Specifically, it was noted that the lift-to-drag ratio (C_L/C_D) is prone to significant degradation under powered-lift conditions due to increased induced drag and viscous mixing in the slipstream.

1.5.1 Revised Design Constraints

To address the anticipated reduction in aerodynamic efficiency during powered flight, the operational parameters were reassessed. It was hypothesized that reducing the operational Angle of Attack (α) for the cruise configuration would assist in recovering the L/D ratio while maintaining sufficient lift margins. Consequently, a secondary comparative study was initiated to evaluate the performance of the shortlisted airfoils (S1221 and S1223) at reduced angles of attack.

1.5.2 Comparative Analysis at Reduced Angles of Attack

The investigation focused on determining if a lower angle of attack could yield the desired aerodynamic efficiency without compromising the high-lift requirements. The following configurations were simulated:

- **Selig S1221:** Evaluated at $\alpha = 12.25^\circ$ and a reduced $\alpha = 8.25^\circ$.
- **Selig S1223:** Evaluated at $\alpha = 13.25^\circ$ and a reduced $\alpha = 9.25^\circ$

1.5.3 Results and Final Decision

The numerical results indicated that the Selig S1223, operating at the reduced angle of attack of $\alpha = 9.25^\circ$, exhibited superior performance characteristics compared to the S1221 configurations.

While the S1221 at $\alpha = 8.25^\circ$ offered improved efficiency compared to its high-AoA state, it failed to match the lift generation capabilities of the S1223. Crucially, the S1223 at $\alpha = 9.25^\circ$ successfully met the target lift-to-drag ratio (C_L/C_D) required for efficient cruise, while simultaneously maintaining a higher lift coefficient (C_L) than the S1221.

Based on these findings, the decision was made to retain the Selig S1223 as the primary airfoil. However, the baseline cruise angle of attack has been formally adjusted from the initially proposed value to 9.25° to optimize the trade-off between power-augmented lift generation and aerodynamic efficiency.

High-Lift Devices

High-lift devices (HLDs) enhance lift during take-off and landing by increasing wing camber and delaying flow separation, but generally incur a drag penalty. This chapter presents a parametric investigation of flap configurations for the selected airfoil, using automated CFD simulations supported by analytical jet-flap modelling to identify configurations that provide high lift with acceptable drag and remain compatible with propulsive lift and three-dimensional wing integration.

2.1 Iterative Flap Configuration Study

2.1.1 Model Description

The S1223 airfoil was used as the baseline section with a **chord length of 0.1 m** and **fixed at an angle of attack of 13.25°** , corresponding to a high-lift take-off condition identified in the airfoil selection study. A single-element flap based on the same airfoil was appended with chord lengths of 20%, 30% and 40% of the main chord. The flap position was parameterized in the streamwise (x) and normal (y) directions relative to the main airfoil trailing edge. Seven x positions with a spacing of 5 mm and five y positions with a spacing of 2 mm were considered. For each position, the flap deflection was varied over five angles in steps of 2.5° about the base orientation. Steady-state CFD simulations were performed for all configurations to evaluate C_L and C_D .

2.1.2 Geometry Parameterisation

A parametric model was developed in **SpaceClaim**, with flap x -position, y -position and deflection angle defined as driving parameters. The computational domain extended 500 mm above and below the airfoils and 1000 mm upstream and downstream. The parametric geometry was linked to ANSYS Workbench to enable automated geometry and mesh updates, with consistent named selections for boundaries and airfoil surfaces.

2.1.3 Grid Independence

Grid convergence was assessed using a representative flap configuration. Several meshes with varying global sizing, trailing-edge refinement and inflation layers were tested. Table below shows representative results.

Beyond Case 2, variations in C_L and C_D were minimal, indicating mesh convergence. A mesh comparable to Case 4 was therefore adopted. The final mesh used a global element size of 4 mm, local refinement near trailing edges, and 18 inflation layers with a first-layer thickness of 4.5×10^{-5} m and a growth rate of 1.17.

Parameter	Case 1	Case 2	Case 4	Case 5	Case 6
Element size (m)	0.012	0.0075	0.004	0.003	0.002
TE edge size (m)	0.0002	0.00012	0.00006	0.00008	0.00006
Airfoil edge size (m)	0.0005	0.0003	0.0002	0.0003	0.0001
First layer thickness (m)	5.75e−05	5.75e−05	4.50e−05	5.70e−05	5.70e−05
Number of layers	8	8	12	7	7
Growth rate (%)	1.2	1.2	1.17	1.2	1.2
AR	12	7.12	6.14	7	2.5
Skewness	0.77	0.675	0.65	0.62	0.66
Element quality	0.15	0.235	0.26	—	—
Number of cells	33,289	265,113	746,916	1,064,982	2,931,274
C_L	2.4603	2.5205	2.5025	2.5029	2.5020
C_D	0.0882	0.0862	0.0869	0.0868	0.0870

TABLE 2.1: Mesh Sensitivity Study and Aerodynamic Results

2.2 Automated CFD Evaluation

2.2.1 Novel Method for batch CFD runs

2.2.1.1 Workbench and Mechanical Scripting

The **Workbench python API of IronPython** was used to automate the geometry updation and mesh generation. Initially, this method was aimed to automate the entire process, however the updation of geometry created outdated mesh zones. The fluent solver was unable to remap these with the older cell zones. From here on the process was split into two parts, mesh generation and solving.

A python script was written to read the geometry parameters from an excel sheet, after which the workbench file was opened and read via the script. The parameters from the excel were assigned to the geometry inputs, and the system was updated. The mesh was initially exported using the *Meshing.ExportMesh* command, however this mesh was incompatible with the fluent solver.

2.2.1.2 PyFluent

For the solution of all the meshes in the fluent environment, **the Ansys python library of PyFluent** was used to automate the process. A case file was setup using the demo mesh, saving all necessary boundary conditions, reference values, turbulence models, and report definitions. They were as follows: PyFluent was opened using a python script and the mesh was replaced one by one from the folder of all meshes. **The solutions ran until residuals of 10e-6 were reached for all cases.** Upon completion of the same, fluent created the report definitions output files which were transferred to a “Reports” folder. **Each solution was also saved as a .dat file in the “Outputs” folder.** Each simulation for a mesh took around 2 minutes, on the same computational setup.

Parameter	Specification 1	Specification 2
Inlet	Velocity Inlet	20m/s
Outlet	Pressure Outlet	0Pa Gauge Pressure
Turbulence Model	Spalart–Allmaras	Turbulence Viscosity Ratio 10
Reference Area	0.1 m ²	
Reference Length	0.1 m	
Report Definitions	Coefficient of Lift (C_L)	Coefficient of Drag (C_D)
Initialization Method	Hybrid Initialization	

TABLE 2.2: CFD Simulation Setup Parameters

2.2.2 Effect of Flap Chord Length

The most promising configuration for a 20% chord flap was extended to 30% and 40% chord lengths. The corresponding aerodynamic performance is summarized in Table.

Flap chord length	C_L	C_D
20% chord	2.4255	0.1568
30% chord	2.3744	0.2012
40% chord	2.4855	0.1901

TABLE 2.3: Aerodynamic coefficients for optimal flap configurations

The 40% chord flap produced the highest lift coefficient, while the 20% chord flap provided lower drag. The range between these values represents a practical design envelope, with the final selection dependent on system-level considerations.

2.2.3 Analysis of heat map results

The analysis of the lift coefficient (C_L) distribution reveals a dominant sensitivity to Geometry Parameter X ($P1$), with the highest performance consistently isolated to the upper bounds of the design space ($P1 \in [0.115, 0.125]$). Across all deflection angles ranging from 0.0° to $+10.0^\circ$, the lift generation remains robust in this region, frequently exceeding a C_L of 2.3. Conversely, Geometry Parameter Y ($P2$) exhibits a less aggressive influence on peak lift, though optimal zones are generally constrained to the lower-to-mid spectrum ($P2 \approx 0.005 - 0.009$). Notably, the heatmaps indicate that increasing the flap deflection angle does not uniformly scale the global maximum lift; rather, it shifts the localized peaks, implying that the geometric gap/overlap parameters ($P1$ and $P2$) are the primary drivers of flow attachment and lift augmentation in this configuration.

In terms of aerodynamic efficiency (C_L/C_D), a distinct inverse correlation is observed with respect to flap deflection angle. The baseline configuration at 0.0° demonstrates superior performance, achieving efficiency ratios nearing 17.0 in the optimal $P1$ region. As the deflection increases to $+10.0^\circ$, the maximum

available efficiency degrades significantly to approximately 10.5, indicating a rapid accumulation of drag that outpaces lift gains. Consequently, the optimal flap position determination should prioritize the geometric coordinates of $P1 \approx 0.120$ and $P2 \approx 0.005$. This specific configuration provides the most robust operational envelope, maintaining peak lift capabilities across all angles while maximizing the efficiency ceiling during lower deflection phases.

The missing values are due to bad geometry generation in the batch run, and thus are omitted from the analysis which is represented by white spaces in heatmaps.

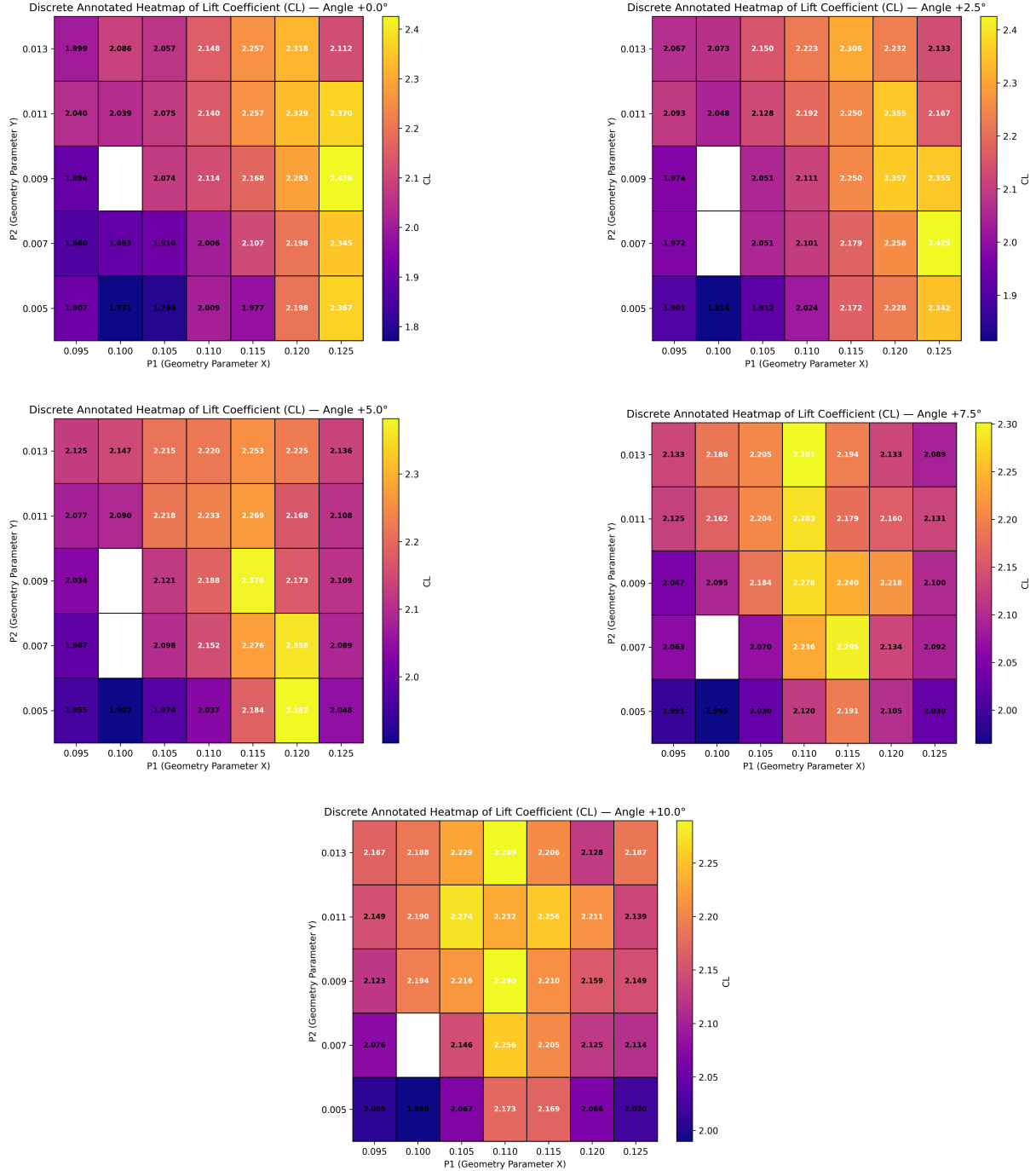
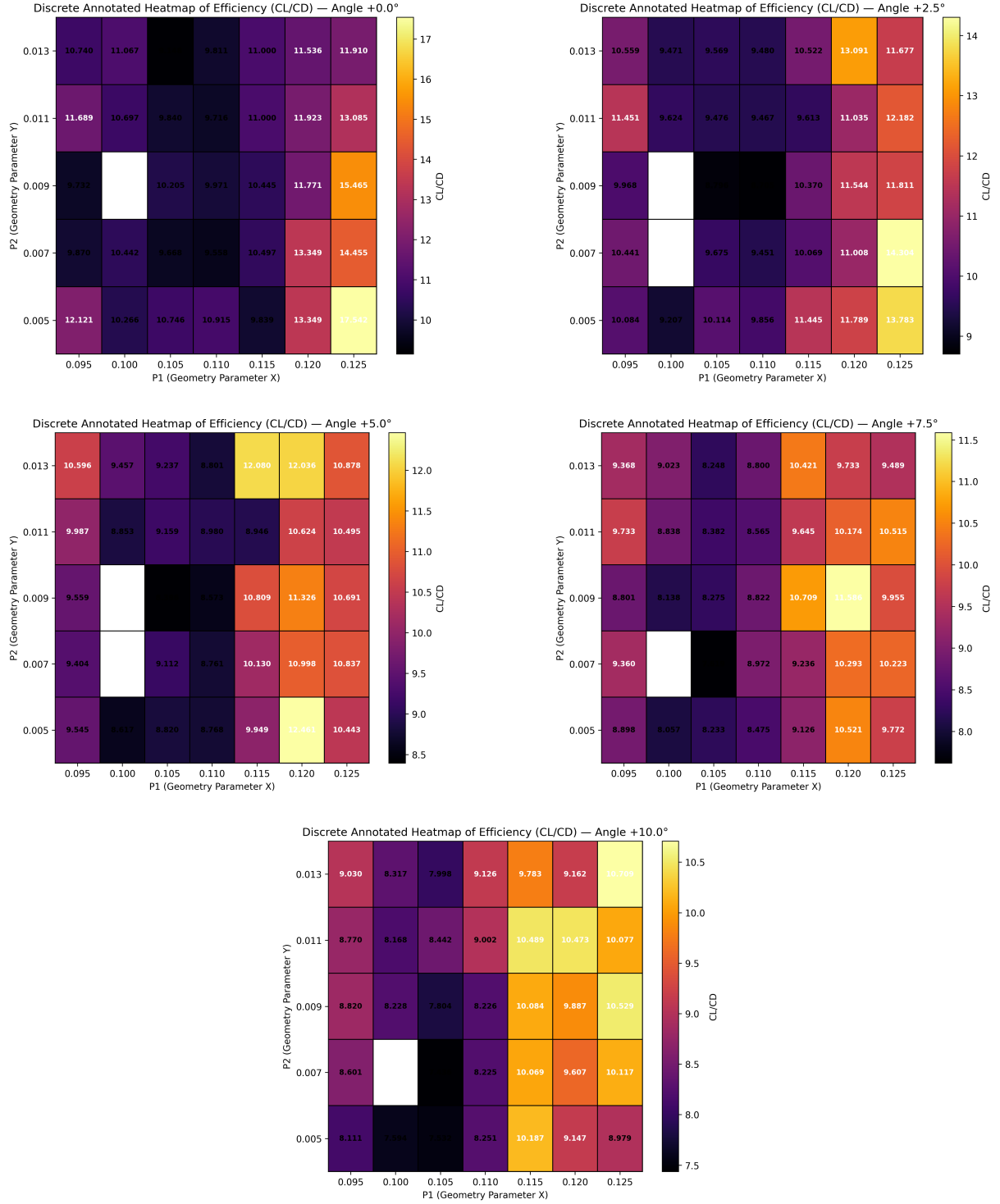


FIGURE 2.1: Heatmaps of CL values for various angles of deflection for the flap

FIGURE 2.2: Heatmaps of C_L/C_D values for various angles of deflection for the flap

The .dat files were used for the computation of the momentum flux coefficient, and for PINN training for further optimization studies.

3-Dimensional Wing Design

3.1 Selection Parameters

The aerodynamic performance of a wing is fundamentally governed by its geometry. While the airfoil shape helps in determining the lift generated in two dimensions, the actual lift of the 3D wing is determined by several parameters, namely aspect ratio, taper ratio, twist angle, sweep angle, and dihedral angle.

3.1.1 Aspect Ratio

Aspect ratio (AR) quantifies the slenderness of the wing planform. It is a primary factor influencing induced drag and lift-curve slope. The aspect ratio is defined as the ratio between the wingspan b and the wing mean aerodynamic chord (MAC) or chord length L_c .

For a finite, unswept wing in incompressible flow, the three-dimensional lift-curve slope a_{3D} and the two-dimensional lift-curve slope a_0 are related through Prandtl's lifting-line theory. Their relation is given by:

$$a_{3D} = \frac{a_0}{1 + \frac{a_0}{\pi e \text{AR}}} \quad (3.1)$$

where e is the Oswald efficiency factor and AR is the aspect ratio.

Reduction in Induced Drag

Since $C_{Di} \propto \frac{1}{\text{AR}}$ (for fixed C_L and e), increasing AR reduces the induced-drag coefficient. Physically, longer (higher-AR) wings reduce the strength of wingtip vortices and associated downwash, making the wing's behavior approach that of a 2-D (infinite-span) airfoil.

Influence on Lift Coefficient

In addition, for a given angle of attack, a wing with higher AR generally exhibits a greater lift-curve slope (i.e. $\frac{dC_L}{d\alpha}$ increases), meaning it can generate higher C_L under similar conditions.

Hence, for a given flight condition (angle of attack, airspeed, air density), a high-AR wing can generate similar or higher lift with reduced induced drag compared to a low-AR wing.

3.1.2 Trade-offs and Practical Considerations

While increasing AR yields aerodynamic benefits, practical and structural trade-offs limit its extent:

- Larger span increases bending moments and structural weight; long wings are heavier and may require stronger (hence heavier) spars and supports.
- More slender wings may increase manufacturing complexity and may restrict maneuver or structural clearance (e.g. hangar space, ground handling).
- Parasitic drag (skin friction, form drag) and structural drag may increase, partly offsetting gains from reduced induced drag.

- The efficiency factor e depends on planform shape and lift distribution; non-ideal planforms (e.g. rectangular wings) have $e < 1$, reducing but not eliminating AR benefit.

3.1.3 Taper Ratio

The taper ratio defines the rate at which the chord decreases from root to tip and significantly influences the spanwise lift distribution, stall characteristics, and overall structural efficiency. It is defined as

$$\lambda = \frac{c_{\text{tip}}}{c_{\text{root}}} \quad (3.2)$$

Typical aerodynamic effects include:

- Moderate taper ($\lambda \approx 0.6$) tends to yield a near-elliptical lift distribution, reducing induced drag.
- Strong taper ($\lambda < 0.3$) can cause very small tip chords and low Reynolds numbers, increasing susceptibility to tip stall.
- Weak taper ($\lambda \approx 1.0$) provides more uniform loading and improved stall behaviour, at the cost of slightly higher induced drag.

To understand the influence of taper ratio on design, a systematic aerodynamic study in XFLR5 using Lifting Line Theory (LLT) under incompressible, subsonic conditions was performed. The wingspan, planform area, and aspect ratio were kept constant to isolate the influence of (λ) . Each configuration was analyzed at a fixed angle of attack of (12.8°) and freestream velocity of 80m/s.

The following taper ratios were evaluated: [$\lambda = 1.00, 0.66, 0.44, 0.25$.]

From the simulation results, the wing with $(\lambda = 0.44)$ produced the spanwise lift distribution closest to the ideal elliptical profile and showed the lowest induced drag, consistent with classical aerodynamic theory [1]. This configuration therefore offered the highest theoretical aerodynamic efficiency among the cases studied.

However, practical design considerations—including manufacturability, structural simplicity, increased torsional rigidity, and improved stall behaviour—favored a uniform chord wing. As a result, despite the aerodynamic benefits of moderate taper, the final design adopts $(\lambda = 1.0)$. This choice balances performance with structural robustness and ease of fabrication while still maintaining acceptable aerodynamic efficiency for the mission requirements.

3.1.4 Calculation of Wing Area and Wingspan

Instead of performing a detailed analytical sizing calculation in this section, the wing geometry was finalised based on the design envelope established from the lift requirements, mission constraints, and aerodynamic targets determined earlier in the study. From this envelope, a **wing planform area of 0.14 m²** and a corresponding **wingspan of 1.4 m for a chord of 0.1m** were selected, as they satisfied the required lift margins for take-off, cruise, and manoeuvring conditions while remaining within the allowable structural and geometric limits of the aircraft.

3.2 No Edge Effect Case

In order to serve as a bridge between 2D and 3D simulations, and also as an additional validation of the 2D CFD, a geometry having a wing edge to edge of the computational domain, which was **100 mm wide** was created. The other dimensions and setup remained the **same**.

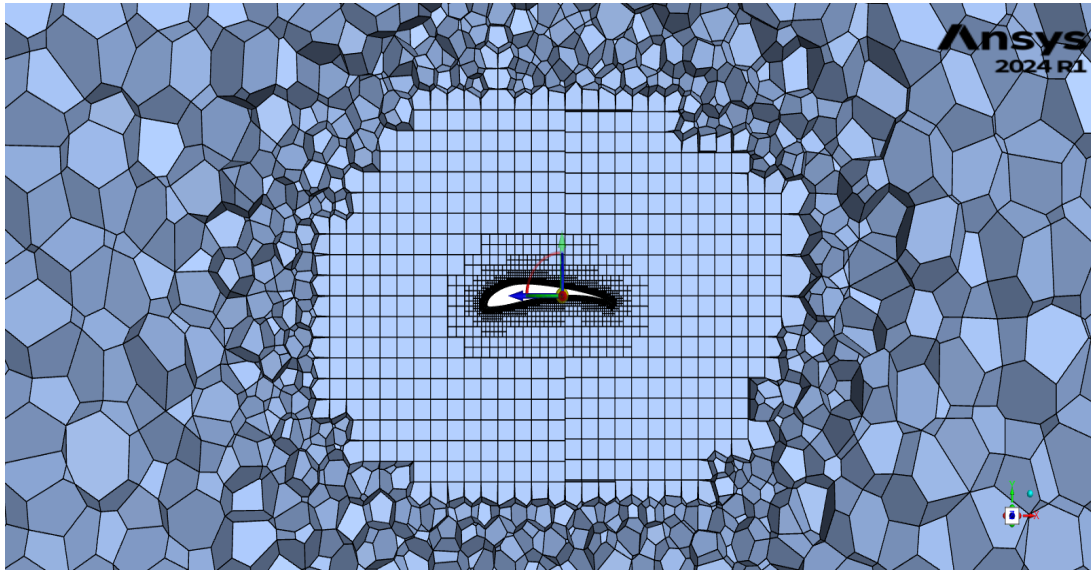


FIGURE 3.1: Finer including inflation layers

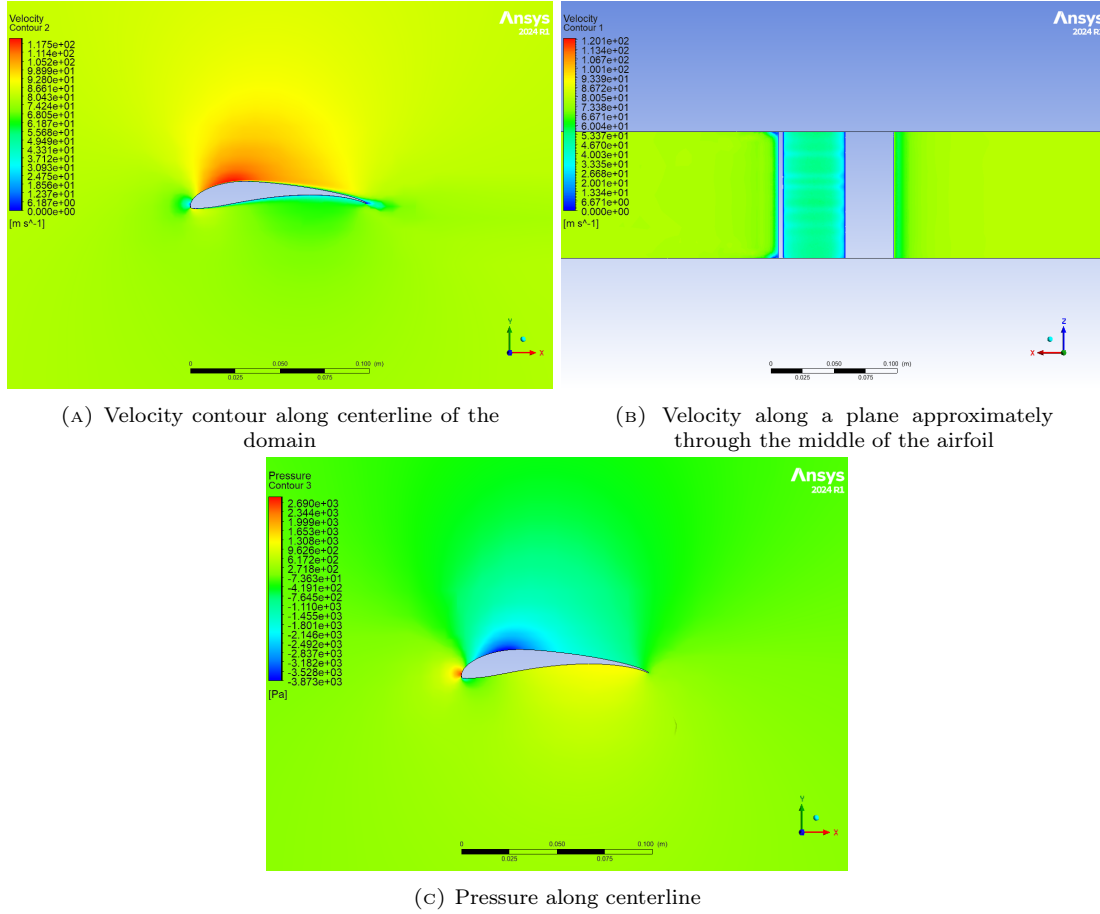


FIGURE 3.2: Velocity and pressure distributions around the airfoil S1223 for semi-2D simulations

3.2.1 Comparative Study Between 2D, No Edge Effect and 3D Simulations

The **2D analysis**, as shown previously, gave $C_L = 1.0008$ and $C_D = 0.02314$ for an angle of attack of 0° for the S1223 airfoil. Considering that 2D analyses should be close to 3D analyses with a depth of 1m, a 3D study with the wing reaching both ends of the domain was created, with the depth being 100mm to reduce computational time. Results were matching closely, with $C_L = 0.9478$ and $C_D = 0.01889$, respectively.

For various mesh sizes, boundary layers, and mesh qualities, the 3D C_L values were consistently in the range of 0.76–0.80. This was evident from the results, as streamlines showed heavy wake generation at the wing tip and its propagation into a larger swirl component. The C_D values increased as well, up to about 0.046, confirming a large increase in the drag created by the wing, by almost a factor of 2.5 compared to the 2D result.

Aerodynamic Center

4.1 Introduction

The aerodynamic center (AC) is a key parameter in the longitudinal stability analysis of aircraft. For a given airfoil or wing section, the aerodynamic center is defined as the point along the chord where the aerodynamic pitching moment coefficient, C_m , is nearly independent of the angle of attack, α . Knowledge of the aerodynamic center is essential in:

- determining static longitudinal stability,
- designing tail surfaces and control systems,
- estimating trim conditions and overall aircraft performance.

The approximate location of the aerodynamic center is first obtained using XFOIL by post-processing the surface pressure and skin-friction distributions for a range of angles of attack. These data are processed in MATLAB to reconstruct the pitching moment coefficient about different reference locations along the chord and to identify the aerodynamic center.

To refine this approximation and remove any dependence on XFOIL modelling assumptions, a separate two-dimensional CFD analysis of the single S1223 airfoil is carried out. The aerodynamic center obtained from this numerical analysis is considered as the final value for the single-element airfoil.

After establishing the aerodynamic center for the single airfoil, the analysis is extended to a three-element high-lift configuration (main airfoil + flap 1 + flap 2) using CFD. The aerodynamic center of the complete configuration is obtained directly from the CFD results and compared with the single-element result.

4.2 Theoretical Background

4.2.1 Definition of the Aerodynamic Center

For a two-dimensional airfoil section, the aerodynamic center is defined as the point along the chord, x_{ac} , where the non-dimensional pitching moment coefficient C_m is approximately independent of the angle of attack:

$$\left. \frac{\partial C_m}{\partial \alpha} \right|_{x=x_{ac}} \approx 0. \quad (4.1)$$

In many practical cases, especially for thin symmetric airfoils at subsonic speeds, the aerodynamic center is close to the quarter-chord point ($x/c \approx 0.25$). However, for cambered and high-lift airfoils, the actual aerodynamic center may deviate from this idealised value, and a more detailed analysis is required.

4.2.2 Pitching Moment Coefficient

The aerodynamic pitching moment about a reference point located at x_{ref} along the chord is defined as

$$M_{x_{\text{ref}}} = \int_0^c p(x) (x - x_{\text{ref}}) dy, \quad (4.2)$$

where $p(x)$ is the pressure distribution and c is the chord length. In practice, for a two-dimensional airfoil, the pitching moment coefficient about a reference point is given by

$$C_{m,x_{\text{ref}}} = \frac{M_{x_{\text{ref}}}}{\frac{1}{2}\rho U_{\infty}^2 c^2}, \quad (4.3)$$

where ρ is the air density and U_{∞} is the free-stream velocity.

If the pitching moment coefficient is known about some reference point x_{ref} , then the pitching moment coefficient about another point x along the chord can be obtained using the standard relation:

$$C_{m,x} = C_{m,x_{\text{ref}}} - C_L \frac{x - x_{\text{ref}}}{c}, \quad (4.4)$$

where C_L is the lift coefficient.

Equation (4.4) is used in this work to reconstruct C_m at various reference locations along the chord, given the lift and moment about a known reference point.

4.3 Single-Element S1223 Airfoil: XFOIL and MATLAB Analysis

4.3.1 Airfoil and Flow Parameters

The baseline airfoil considered in this study is the S1223 airfoil. The flow conditions and analysis parameters used in XFOIL are:

- Airfoil: S1223,
- Angle-of-attack range: $\alpha = 0^\circ$ to 5° with increments of 0.1° ,
- Reynolds number: $Re = 137,000$,
- Mach number: $M = 0.038$,
- Reference chord length: c (non-dimensionalised such that $x/c \in [0, 1]$).

4.3.2 XFOIL Data Generation

XFOIL is used to compute the aerodynamic characteristics of the S1223 airfoil over the specified range of angles of attack. For each angle of attack, the following quantities are extracted:

- pressure coefficient distribution $C_p(x)$ over the airfoil surface,
- skin-friction coefficient distribution $C_f(x)$,
- integrated aerodynamic coefficients such as lift coefficient C_L and pitching moment coefficient about a default reference point (typically quarter-chord or leading edge), if available.

Direct C_m data was not taken from XFOIL. Instead, the surface distributions C_p and C_f to reconstruct the aerodynamic loads and corresponding moments about different reference locations along the chord.

4.3.3 Post-Processing in MATLAB

All data exported from XFOIL (pressure and skin-friction distributions for each α) are imported into MATLAB. The algorithm implemented in MATLAB performs the following key steps:

1. Read $C_p(x)$ and $C_f(x)$ for each angle of attack in the range 0° – 5° with $\Delta\alpha = 0.1^\circ$.
2. Compute the local pressure forces on each surface panel and integrate them to obtain:
 - sectional lift coefficient C_L ,
 - sectional drag coefficient C_D ,
 - pitching moment coefficient about a chosen reference point $C_{m,x_{\text{ref}}}$.
3. Use the moment-shift relation in Eq. (4.4) to evaluate the pitching moment coefficient for a series of reference locations x/c in the interval $[0, 1]$ with a small step $\Delta(x/c) = 0.001$.
4. For each reference location x/c , construct the curve $C_m(\alpha)$ and examine its variation with angle of attack.

The algorithm is designed to systematically scan along the chord and identify the location where C_m is least sensitive to α .

4.3.4 Aerodynamic Center from XFOIL + MATLAB

For each candidate reference location x/c , the variation of C_m with α is analysed. The aerodynamic center is approximated as the location where:

- the curves of C_m versus x_{ref}/c for different values of α intersect or nearly intersect, and/or
- the slope $\partial C_m / \partial \alpha$ is minimised in a least-squares sense.

From the MATLAB-generated results, plots of C_m as a function of x_{ref}/c for several angles of attack were created. The point where these curves nearly intersect is taken as the aerodynamic center. For the S1223 airfoil, the aerodynamic center was found to be approximately

$$\frac{x_{\text{ac, single, XFOIL}}}{c} \approx 0.247. \quad (4.5)$$

The plot used to identify the aerodynamic center is shown in Figure 4.1.

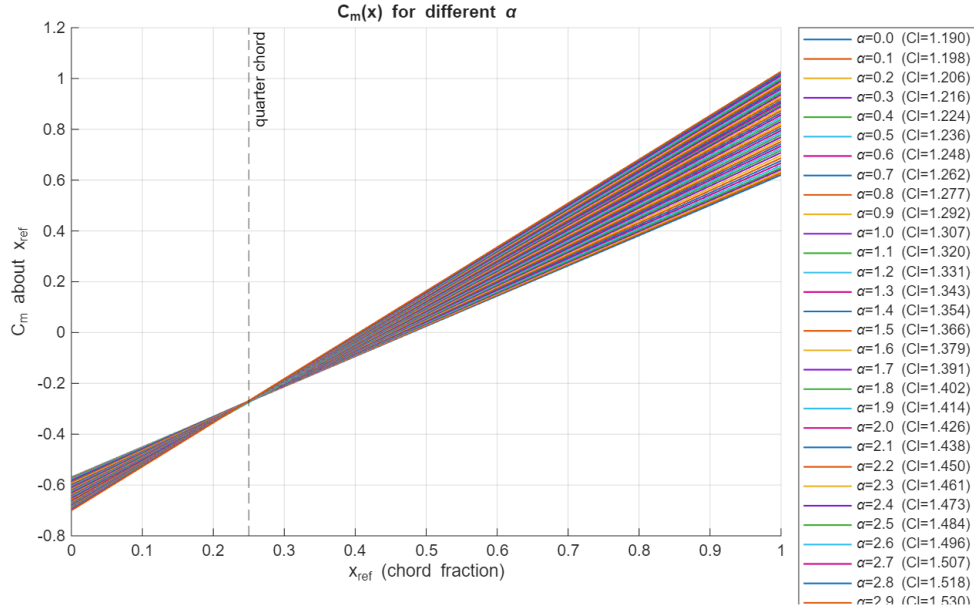


FIGURE 4.1: Pitching moment coefficient C_m as a function of reference location x_{ref}/c for different angles of attack for the S1223 airfoil (XFOIL + MATLAB). The intersection region of the curves indicates the aerodynamic center.

The variation of C_m with α evaluated at the aerodynamic center location is shown in Figure 4.2. The near-constant behaviour confirms that the identified position is a good approximation of the aerodynamic center for the single S1223 airfoil.

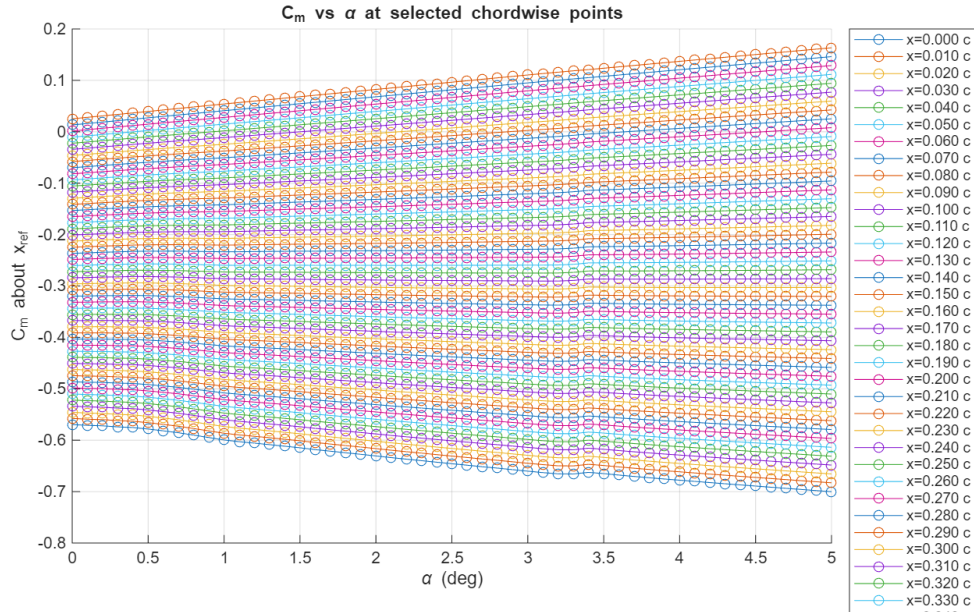


FIGURE 4.2: Pitching moment coefficient C_m versus angle of attack α for the S1223 airfoil evaluated at the identified aerodynamic center (XFOIL + MATLAB).

4.4 Single-Element S1223 Airfoil: CFD Numerical Analysis

4.4.1 CFD Setup for Single Airfoil

To obtain a more precise and fully numerical estimate of the aerodynamic center, a two-dimensional CFD simulation of the single S1223 airfoil is performed in ANSYS Fluent. The flow conditions are chosen to be consistent with the XFOIL analysis:

- same Reynolds number and Mach number as the XFOIL study,
- incompressible, steady-state flow,
- Spalart–Allmaras turbulence model,
- free-stream velocity corresponding to $U_\infty = 20\text{m/s}$,
- far-field boundaries sufficiently far away from the airfoil,
- no-slip boundary condition on the airfoil surface.

4.4.2 Extraction of Single-Airfoil CFD Aerodynamic Center

For each angle of attack considered in the CFD simulation, the pressure and wall-shear stress distributions are integrated over the airfoil surface to obtain C_L , C_D and the pitching moment coefficient about a convenient reference point (for example, the leading edge). Using Eq. (4.4), the pitching moment coefficient is then evaluated at different reference locations x/c near the quarter-chord region.

The family of curves $C_m(x_{\text{ref}}/c)$ for different α is constructed, and the aerodynamic center is again identified as the location where C_m is almost independent of α . The corresponding results are summarised in Figure 4.3, where the curves obtained from CFD are shown.

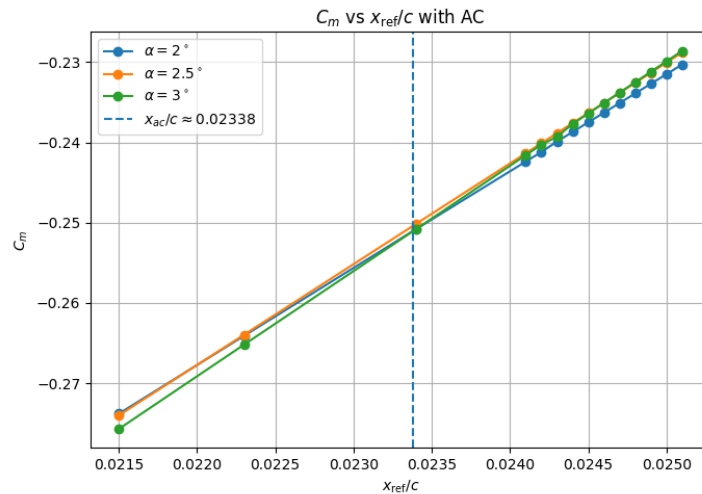


FIGURE 4.3: Pitching moment coefficient C_m as a function of reference location x_{ref}/c for different angles of attack for the single S1223 airfoil (CFD).

From this CFD-based analysis, the aerodynamic center of the single S1223 airfoil is obtained as

$$\frac{x_{\text{ac, single, CFD}}}{c} \approx 0.23381, \quad (4.6)$$

which is used as the final and most accurate estimate for the single-airfoil aerodynamic center in this work. The corresponding variation of C_m with α at this location is shown in Figure 4.4.

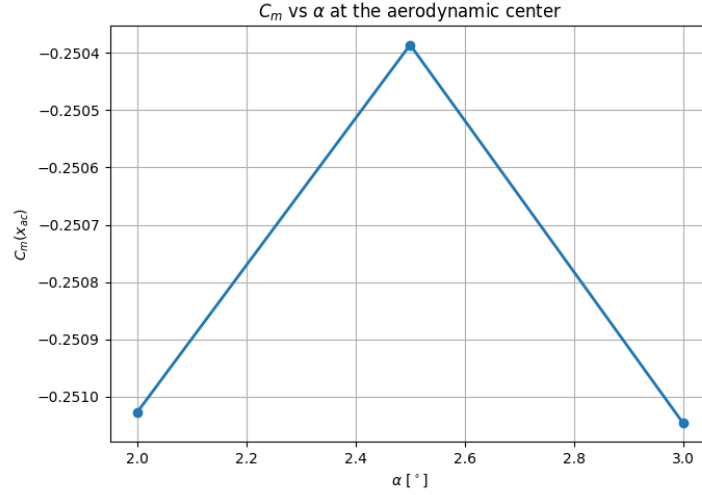


FIGURE 4.4: Pitching moment coefficient C_m versus angle of attack α at the CFD-based aerodynamic center of the single S1223 airfoil.

The CFD result shows good agreement with the XFOIL + MATLAB estimate and provides a more reliable value since it fully resolves the flow field around the airfoil.

4.5 Three-Element High-Lift Configuration: CFD Methodology

4.5.1 Geometry and Configuration

After establishing the aerodynamic center of the single S1223 airfoil, the analysis is extended to a three-element high-lift configuration. The configuration consists of:

- Main airfoil: S1223,
- Flap 1: chord length $0.4c$ (40% of the main airfoil chord) with a specified deflection and position relative to the main element,
- Flap 2: chord length $0.2c$ (20% of the main airfoil chord) with a specified deflection and position relative to the main element.

4.5.2 Computational Domain and Boundary Conditions

The CFD simulations for the three-element configuration are carried out using ANSYS Fluent. The key features of the computational setup are:

- two-dimensional steady-state simulation,
- incompressible, low Mach number flow,
- turbulence model: Spalart–Allmaras,
- inlet boundary: uniform velocity corresponding to free-stream speed $U_\infty = 20\text{ m/s}$,
- outlet boundary: pressure outlet with ambient static pressure,
- far-field boundaries placed sufficiently far from the airfoil to minimise blockage effects,

- no-slip boundary condition on all solid surfaces (main airfoil and flaps).

4.5.3 Extraction of Aerodynamic Coefficients

Once convergence is achieved for each flow condition (angle of attack), the pressure and shear stress distributions are integrated over all three elements to obtain:

- total lift coefficient C_L of the three-element system,
- total drag coefficient C_D ,
- pitching moment coefficient $C_{m,x_{\text{ref}}}$ about a chosen reference point (for example, the leading edge of the main element).

Using Eq. (4.4), the pitching moment coefficient is then calculated for a range of reference locations x/c near the approximate aerodynamic center obtained from the single-element CFD study. This procedure is repeated for several angles of attack.

4.6 Three-Element Configuration: CFD Results

For the three-element configuration, CFD provides the total aerodynamic coefficients as well as the pitching moment about the chosen reference point.

Figure 4.5 shows the variation of C_m with x_{ref}/c for multiple angles of attack for the three-element system. The intersection region of these curves corresponds to the aerodynamic center of the combined configuration.

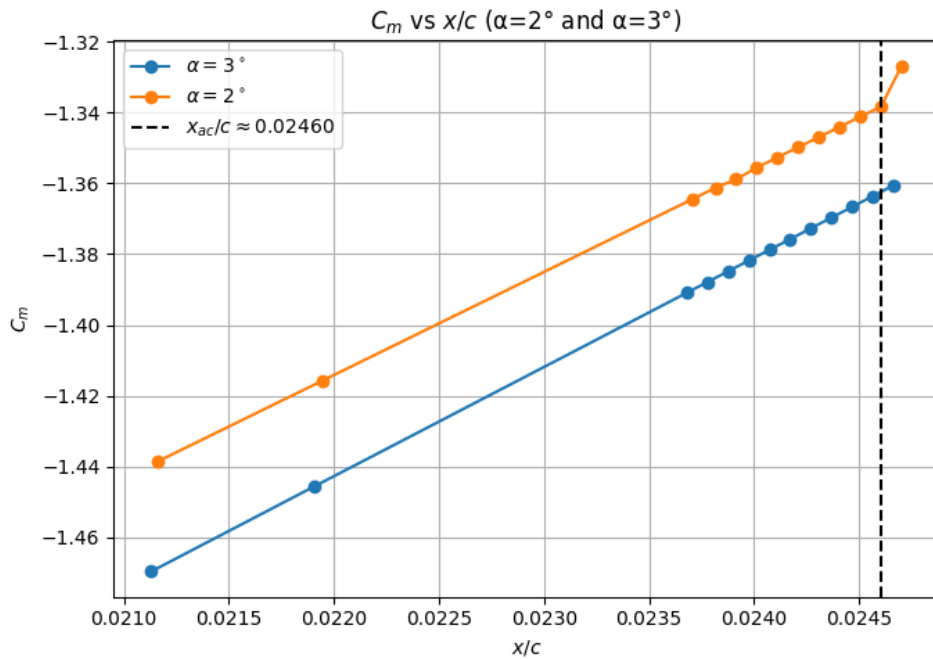


FIGURE 4.5: Pitching moment coefficient C_m as a function of reference location x_{ref}/c for different angles of attack for the three-element configuration (CFD).

The aerodynamic center of the three-element configuration is found to be:

$$\frac{x_{ac,three}}{c} \approx 0.24603, \quad (4.7)$$

4.7 Comparison and Discussion

A comparison of the aerodynamic center locations for the different configurations and methods is summarised in Table 4.1.

Configuration	Method	x_{ac}/c
Single-element S1223	XFOIL + MATLAB	0.247
Single-element S1223	CFD	0.23381
Three-element (S1223 + flap 1 + flap 2)	CFD	0.24603

TABLE 4.1: Comparison of aerodynamic center locations for single-element S1223 and three-element configuration

The CFD-based aerodynamic center for the single S1223 airfoil shows good agreement with the XFOIL + MATLAB prediction and is adopted as the final value because it is obtained from a full solution of the Navier–Stokes equations, with no panel-method assumptions. The shift in the aerodynamic center from the single-airfoil CFD value to the three-element configuration can be explained by:

- increased camber and circulation due to the deflected flaps,
- redistribution of pressure and lift across the chord and over multiple elements,
- additional nose-down or nose-up pitching moment contributions from the flaps depending on their deflection.

Overall, the CFD-based aerodynamic center for the three-element configuration lies in the vicinity of, but not identical to, the value for the single-element S1223 airfoil. This demonstrates the importance of performing a dedicated analysis for multi-element high-lift systems rather than relying solely on the single-element airfoil behaviour.

Propulsion

5.1 Introduction

This system consists of the following components - the power source, the power generation devices, and the rotors. As far as the power source and power generation devices are concerned, we are assuming that the powertrain altogether is sufficient to provide realistic values of power to the rotors, whose study has been conducted below.

The rotors serve the purpose of generating thrust out of shaft power supplied to it via the power generation devices. The thrust generation done by the rotors are used to propel the aircraft in the horizontal direction and make it reach speeds at which the lift generated by the wings are enough for the aircraft to takeoff. However, in the case of STOL (Short Take-Off and Landing) aircrafts, this high-velocity air being generated by the rotors is often utilised for re-energizing the boundary layer of the flow on the main wing: hence delaying boundary layer separation and allowing for higher attack angles and ultimately, higher lift.

5.2 Propulsion Types

Between open propeller systems (or open rotors) and Electric Ducted Fans (EDFs), the comparison centres around efficiency and slipstream characteristics. Open propellers are well known for their superior propulsive efficiency at low and medium airspeeds, mainly due to their large disc area and reduced aerodynamic losses. They generate a wide, relatively uniform slipstream that effectively energises the airflow over the wing, improving lift and delaying flow separation. In contrast, EDFs are generally unsuitable for sustained high-speed cruise due to their low momentum and propulsive efficiency, compounded by fan and duct aerodynamic losses. At elevated flight velocities they also require much higher rotational speeds to maintain the necessary pressure rise, leading to increased power consumption and higher mechanical and thermal loads compared to a properly sized open propeller. Hence, the open-prop configuration was selected for this study, as it provides a broad and steady slipstream distribution over the wing, **ensuring smoother flow behaviour and more uniform lift enhancement.** This characteristic directly supports the objective of achieving high lift coefficients for the powered-lift wing design. Moreover, the open propeller achieves these benefits without excessive power consumption or complex mechanical actuation, making it the most feasible and practical choice for a hybrid-electric STOL platform aimed at maximising lift performance under realistic operating constraints.

5.3 Distributed Propulsion Systems

The most efficient and high-lift producing wings have the innate characteristics of having uniform airflow over the wings. Consider a twin engine jet or turboprop, the part of the wing within the wake of the propulsive device will have a higher mass flow rate than the other areas, and hence a higher value of C_L . **For the purpose of having propulsive lift, the propulsive device “blows” air over the wing in order to increase the airflow and improve characteristics.** In such a scenario, having airflow uniformly distributed over the wingspan greatly improves wing performance, both aerodynamically and structurally. Hence arises the need and the term, distributed propulsion.

Distributed propulsion is essentially discretizing the propulsion over the wingspan, in the form of multiple propulsive devices. This can be done in some different ways, all having inherent advantages and disadvantages. Perhaps the most intuitive is distributing the thrust requirement equally over the wingspan by having multiple propulsive devices, deciding the wingspan and size of propulsive device accordingly. Another possible approach might be to have multiple smaller propulsive systems, for the sole purpose of boosting airflow over the wing, and having one or more main propulsive devices, for meeting the thrust envelope requirements.

For the particular application of STOL aircrafts, aspects like aerodynamic efficiency were considered. Looking at the possible scenarios, 4 propellers of 20cm diameter were evaluated, having a hybrid approach of approximately 80% wing area coverage for an AR of 10. Other options include 10 propellers, 5 on each side with approximately 10cm. Further analysis and evaluation would be done, based on the developed and validated methods of CFD and analytical models.

5.4 Blade Element Theory

5.4.1 Introduction to Blade Element Momentum Method

The Blade Element Moment Method originates from momentum theory (MT) and the Blade Element Theory (BET). The BEM algorithm can be used to determine the required twist angles and the Overall performance of the Propeller if the known parameters: free stream velocity (V_{inf}), rotational speed (ω), Number of Blades (B), etc. **Most BEM implementations, including the one from Glauret, rely on the streamtube theory** version of the momentum theory, so that the different radial positions are assumed to be independent. The streamtube momentum theory is applied to an elementary annulus of radius r , providing elementary Thrust and Torque dT and dQ for calculated induction factors a and a' .

5.4.2 Linear momentum

Using Mass balance equation

$$\rho A_\infty U_\infty = \rho A_d U_d = \rho A_w A_w \quad (5.1)$$

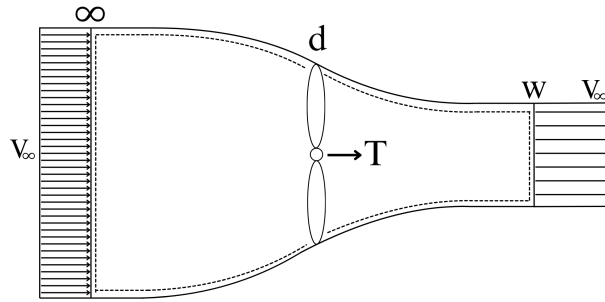


FIGURE 5.1: Control Volume 1

Using Momentum Balance equation

$$\rho V_{\infty}(-V_{\infty}A_{\infty}) + \rho V_w(V_w A_w) = T \quad (5.2)$$

where,

- ρ = Density of Air (assuming that air is incompressible)
- A = Cross-sectional area
- U = Velocity at a section
- T = Thrust

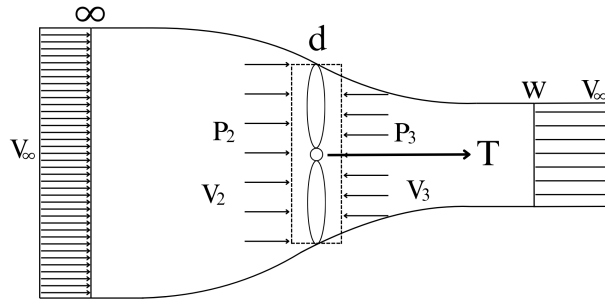


FIGURE 5.2: Control Volume 2

Momentum balance equation in control volume 2

$$\rho V_2(-V_2 A) + \rho V_3(V_3 A) = T + (P_2 - P_3)A \quad (5.3)$$

from equations (1), (2) & (3)

$$\rho V_d(V_w - V_{\infty}) = P_3 - P_2 \quad (5.4)$$

Using Bernoulli's equation from ∞ to 2 and 3 to w

Where, u = Velocity induced just behind the propeller.

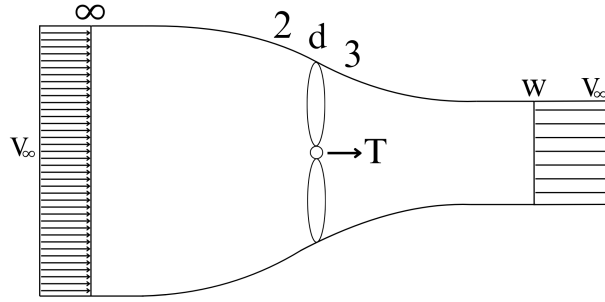


FIGURE 5.3: Application of Bernoulli's Theorem on Control volume

therefore,

$$V_d = V_\infty(1 + a) \quad (5.5)$$

where, $a = \frac{u}{V_\infty}$ and is called axial induction factor.

By combining all the equations, the thrust equation is achieved for a section.

$$T' = 2a(1 + a)\rho V_\infty A \quad (5.6)$$

5.4.3 Angular Momentum

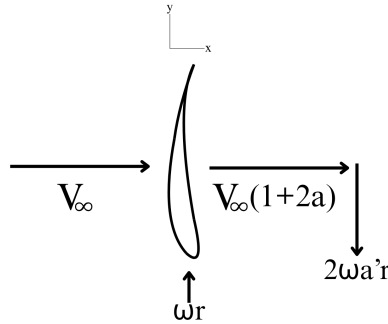


FIGURE 5.4: Cross section of Blade

$$\text{Total Velocity} = \omega r(1 - a') \quad (5.7)$$

where, $a' = V_{swirl}/\omega r$ is called the tangential induction factor

By the Momentum equation

$$r \times \left[\frac{\partial}{\partial t} \int \rho V dV + \int \rho V (V \cdot A) \right] = r \times F \quad (5.8)$$

$$Q' = 4a'(1 + a) \rho U_\infty \omega r^2 A \quad (5.9)$$

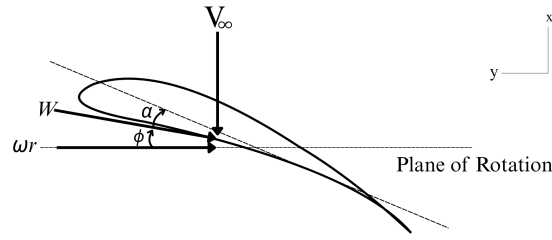


FIGURE 5.5: Velocity components and angle definitions at a blade element

5.4.4 Blade Element Theory

α : Angle of attack of the airfoil ϕ : Inflow angle θ : Twist Angle.

$$\alpha = \theta - \phi \quad (5.10)$$

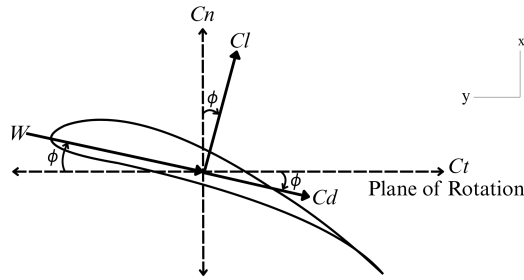


FIGURE 5.6: Resolution of aerodynamic forces on a blade element

$$C_n = C_l \cdot \cos(\phi) - C_d \cdot \sin(\phi) \quad (5.11)$$

$$C_t = C_l \cdot \sin(\phi) + C_d \cdot \cos(\phi) \quad (5.12)$$

$$T' = \frac{1}{2} C_n \rho W^2 BC \quad (5.13)$$

$$Q' = \frac{1}{2} C_t \rho r W^2 BC \quad (5.14)$$

from equation (5.6) and (5.13)

$$a = \frac{1}{\frac{4F \sin^2(\phi)}{\sigma' C_n} - 1} \quad (5.15)$$

from equation (5.9) and (5.14)

$$a' = \frac{1}{\frac{4F \sin(\phi) \cos(\phi)}{\sigma' C_t} + 1} \quad (5.16)$$

where,

- $\sigma' = (\frac{BC}{2\pi r})$ is called local solidity.
- F = Tip loss coefficient.

5.4.4.1 Tip loss Corrections

The solution approach utilises tip loss correction approximations to mitigate errors in both theoretical and practical predictions. So, the outputs of the solution can be taken into consideration without taking excessive buffers to accommodate the losses.

The solution utilises classical Prandtl tip loss correction functions to evaluate loss correction factors. Prandtl tip loss function forces circulation (thus, the lift) to go to zero at blade tips and root.

Solution has made use of commonly used formulae for tip loss correction factor and root loss correction factors,

$$F_{tip} = \frac{2}{\pi} \cos^{-1} \left[\exp\left(\frac{-B \cdot (R_{blade} - r)}{2r \cdot \sin\phi}\right) \right] \quad (5.17)$$

$$F_{root} = \frac{2}{\pi} \cos^{-1} \left[\exp\left(\frac{-B \cdot (r - R_{hub})}{2R_{hub} \cdot \sin\phi}\right) \right] \quad (5.18)$$

$$F = F_{tip} \times F_{root} \quad (5.19)$$

where,

- B = number of blades on the propeller.
- ϕ = Inflow angle.
- F_{tip} = Tip loss correction factor.
- F_{root} = Hub or root loss correction factor.
- F = Overall tip loss correction factor.

5.4.5 Solution approach

From fig. 5.6

$$\tan(\phi) = \frac{V_{\infty}(1+a)}{wr(1-a')} \quad (5.20)$$

from equation (22) a residual function can be formed

$$R(\phi) = \frac{\sin(\phi)}{1+a} - \frac{V_{\infty} \cos(\phi)}{wr(1-a')} ; \quad \phi = (0, \frac{\pi}{2}] \quad (5.21)$$

to obtain the optimum values of the residual function ϕ must approach zero or be within the decided tolerance value.

5.5 Simulation Framework

5.5.1 Methodology

- **Importing the Airfoil geometry using the .dat file.** This loaded a dimensionless geometry of the cross section of the blade to be designed later. For the selection of the airfoil for the rotor, we

again conducted a study and found S1223 - high lift low Reynolds number airfoil to be the most appropriate for the reasons listed below:

- **High Lift Coefficient** even at low RPMs (low Reynolds Numbers), directly improving static thrust and low speed lift generation.
 - **High CL/CD ratios** at low Angle of Attack (AoA) implying a good propulsive efficiency.
 - Stall angle distributed over a range, rather than a sudden stall, offering redundancy in case of change in inflow angles due to changes in the operating conditions.
 - **Thickness (12%) and Camber (8%) such that the airfoil is both structurally sound while being sufficient in terms of lift production.**
 - Already used in a variety of small production propellers.
- Performing XFOIL analysis on the airfoil for a range of Angle of Attacks, for various Reynolds Numbers, by defining different Partial Polars. A Polar refers to a variable (here, either C_L , C_D , C_m) tabulated over a range of AoAs. These polars are said to be ‘partial’ since the range of AOAs is a subset of the range -180° to $+180^\circ$. For the case under discussion, a range of Reynolds numbers equal to 30,000 to 120,000, with intervals of 10,000, was used to make multiple partial polars, upon which analysis was run.
 - **Extrapolation of partial polars to 360° or Complete Polars using the inbuilt extrapolation methods inside QBlade.** It often occurs that as a result of the geometry, operational state or other factors that AoA values outside of this range (that the partial polars are in), are experienced by the airfoils of the blade. For this reason, the blade creation module of QBlade requires that polars are defined for the full 360° AoA range. And within this process, there are two ways possible to do the extrapolation:
 - **Viterna Extrapolation:** The first option available to extrapolate airfoil polar data is the Viterna method. The generated extrapolation is visualized in the graph section to allow for optimization. The following parameters can be tuned to improve the behavior of the polar - Range of the original polar, C_{D90} (which specifies the value of C_D at 90° , which influences the lift coefficient behaviour) and $St+/St-$ (which specify the positive and negative stall AoA for the airfoil, respectively.)
 - **Montgomery Extrapolation: This method is based on the assumption that the airfoil acts aerodynamically as a flat plate for high values of AoA.** The generated extrapolation is visualized in the graph section to allow for optimization. The following parameters can be tuned to improve the behavior of the polar - $A+/B+$, $A-/B-$ (curve parameters for the positive and negative side of the AoA range, respectively), **Slope, and C_{D90} .**
 - **Inputting the blade geometry being output from the mathematical model.** This involved entering the values of chord, twist, airfoil and Reynolds number range corresponding to each

discretisation of the blade. This resulted in a 3D Model of the blade geometry that is generated from the mathematical model for its inputs.

- **Running BEM Simulations at conditions identical to that of the mathematical model.**

QBlade allows the user to run simulations in two modes: Prop BEM and Multi BEM. For this study, Prop BEM Mode was utilised, and the simulations were run at fixed conditions in terms of RPM, Collective Pitch, Density, Kinematic Viscosity, Number of Discretizations of the blade, Maximum Epsilon for Convergence, Maximum Number of Iterations and Relax Factors. Moreover, all simulations were run without Prandtl Tip Loss and 3D corrections in order to be consistent with the mathematical model.

Category	Parameter	Value / Description
Operating Condition	Propeller RPM	14000 rpm
Blade Setting	Collective Pitch	0°
Fluid Properties	Density	1.225 kg/m ³
	Kinematic Viscosity	1.79×10^{-5} m ² /s
Numerical Setup	Blade Discretization	1000 elements
Numerical Convergence	Max Epsilon	1×10^{-6}
	Max Iterations	1000
	Relaxation Factor	0.1
Corrections	Prandtl Tip Loss	Not applied
	3D Correction	Not applied

TABLE 5.1: BEM Parameters taken into account for simulations

Comparison was carried out at advance ratios as calculated from the geometry.

5.5.2 Comparative study

As discussed earlier, all simulations were run without Prandtl Tip Loss and 3D corrections in order to be consistent with the mathematical model. Moreover, the simulations were run at fixed conditions in terms of RPM (14000), Density (1.225 kg/m³), Kinematic Viscosity (1.79e-05), Number of Discretizations of the blade (1000), Maximum Epsilon for Convergence (1e-06), Maximum Number of Iterations (1000) and Relax Factor (0.1). The following table compares some of the test cases used to verify thrust predictions from the mathematical model.

Case No.	No. of Blades	Blade Length (m)	Rotor Diameter (m)	Re _{min}	Re _{max}	Thrust (N) — Math Model	Thrust (N) — QBlade	% Error
1	5	0.09	0.20	30,000	110,000	43.07	47.21	8.77
2	4	0.09	0.20	30,000	110,000	34.31	36.42	5.80
3	3	0.09	0.20	30,000	110,000	31.10	27.85	11.67

TABLE 5.2: Comparison of Thrust Prediction between Mathematical Model and QBlade

5.6 Application of BEM

5.6.1 Code

The team has developed a MATLAB script that generates the most efficient geometry according to the Blade Element Momentum Theory (BEMT) and given inputs.

5.6.1.1 Main Script

1. Setup Constants

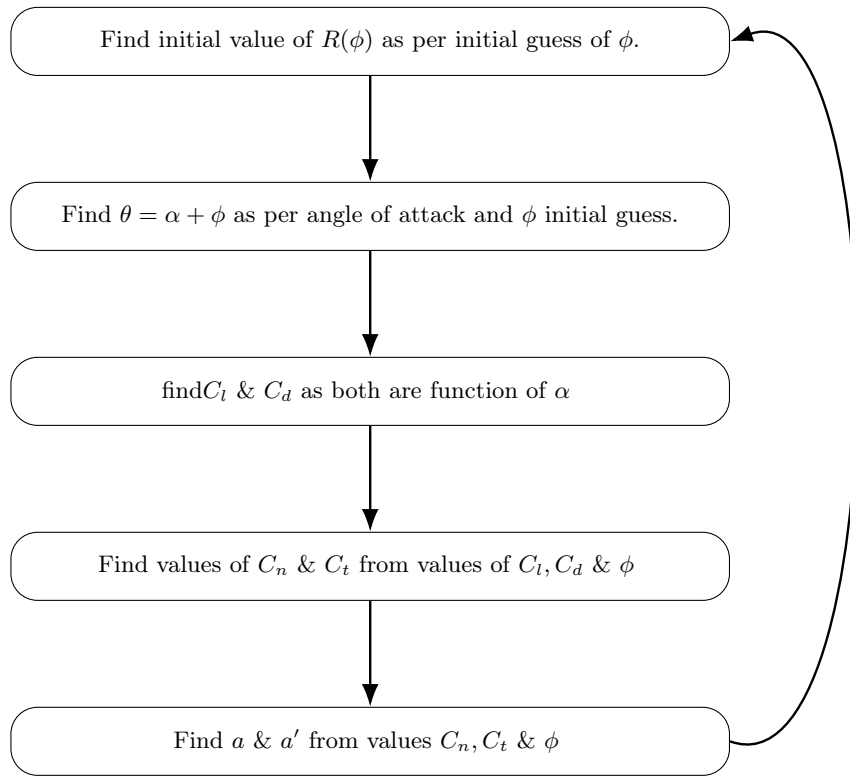
- V_{inf} , ω , B , R , ρ , etc. are input parameters like freestream velocity, RPM, number of blades, radius, air density, etc.
- r is the radius of a section on a blade.
- a and ap are initial guesses of axial induction factor and tangential induction factor to start iterations.

2. Loading airfoil data (*S1223_polars.mat*)

- Contains values of C_L and C_D vs Angle of attack (α) for different Reynolds numbers.
- Once loaded *Blade_designer.mat* can be run.

3. Main loop (for each blade element $r(i)$)

- Starts with initial guesses of a , ap and ϕ .
- Iteratively solves for the correct flow angle(ϕ) and induced velocities until convergence.
- In each iteration:
 - (a) Compute axial and tangential velocity (V_a and V_t)
 - (b) Compute Reynolds number for that section.
 - (c) Calculates the best AoA ($Alpha_best$) from *getBestAlpha.m*.
 - (d) Sets twist (θ) so that local AoA = Alphabest.
 - (e) Interpolates to find C_L and C_D from airfoil data using *interpolatePolar.m*
 - (f) Computes normal (C_n) and tangential (C_t) force coefficients.
 - (g) Computes Prandtl tip/root loss factor (F) if enabled.
 - (h) Computes new a and ap Using momentum equations (Uses Glauert correction for high thrust.)
 - (i) Relaxes the update (to avoid oscillations).
 - (j) Repeats until solution converges.

FIGURE 5.7: Flowchart for iterative computation of ϕ and $R(\phi)$

4. After Convergence

- Computes sectional thrust ($\frac{dT}{dr}$) and torque ($\frac{dQ}{dr}$) using momentum theory.
- Integrates both along the blade to get the total values.
- Prints both and makes a table (T) summarising both

5.6.1.2 Polar interpolater (*interpolatePolar.m*)

- This helper function finds C_L and C_D for a given angle of attack and Reynolds number by interpolating between the nearest tables.

- Steps:

1. Clamps Re within available data range.
2. Finds the two closest Reynolds number tables (low and high).
3. Extracts their C_L and C_D vs α data.
4. Interpolates within each table to get

$$\frac{C_L}{C_D}$$

a given α .

5. Linearly blends between both tables depending on Re .
6. Gives the final interpolated C_L and C_D .

5.6.1.3 Polar Generator Script (*S1223_polars.mat*)

This script creates the *S1223_polars.mat* file from your XFOIL CSVs.

Steps:

1. Defines all Res (*Re_vals*) and corresponding CSV file names
2. Loop through all CSVs:
 - (a) Reads them with *readtable()*.
 - (b) Extracts α , C_L , and C_D columns (case-sensitive)
 - (c) Removes bad rows, sort by α .
 - (d) Stores each as a table inside a cell array *polar_data*.
3. Saves everything to a .mat file for use in the main propeller code.

5.6.1.4 String to double converter (*toNumericColumn.m*)

Utility function that converts messy text columns to clean numeric arrays.

Steps:

1. If already numeric, just returns it.
2. Otherwise,
 - (a) Replaces commas with periods (for decimals).
 - (b) Extracts numeric text from the string.
 - (c) converts it to *double*.
3. Returns clean numeric vector.

5.7 CFD of propeller

The geometry is created via values from the analytical code, using a CAD software. **The mesh is generated using ICEM CFD, containing a rotating region housing the propeller, and an enclosure, having 5D as the entrance length, and 15D as the downstream length, and 5D on each side of the rotating region vertically (D is the diameter of propeller).**

The turbulence model used will be K- ω -SST to accurately model the tip losses and swirl. The inlet velocities simulated are 20m/s and 80m/s. The solver which will be used is ANSYS CFX.

Due to the geometric complexity and the large domain size, the mesh required an exceptionally high number of elements, resulting in substantial computational demands. Therefore, the team adopted alternative methods as outlined below.

5.8 Actuator Disk Theory

Actuator Disk Theory (ADT) is used for analysing a rotor such as a propeller or turbine by replacing its blades with a thin permeable disk. The disk allows the flow to pass and simultaneously acts as a momentum source to the fluid. This enables the aerodynamic effect of a propeller to be incorporated

without physically modelling its blades, thereby reducing the computational requirements significantly. ADT is based on the conservation of momentum, mass, and energy.

Properties of the actuator disk:

- The flow is a perfect fluid, steady and incompressible.
- The actuator disk represents the propeller blades and provides energy to the flow.
- The momentum source creates a pressure discontinuity across the disk.

Assumptions related to ADT:

- The Mach number is low, ensuring incompressible flow.
- Flow outside the propeller streamtube maintains constant stagnation pressure.
- The rotating blades are smeared into a single, thin, steady disk.
- Pressure changes discontinuously across the disk, but velocity remains continuous.
- The axial induction factor a allows wake velocity to be quantified as:

$$U_{\text{wake}} = (1 + 2a)U_{\infty} \quad (5.22)$$

where U_{∞} is the upstream velocity at the turbine hub height.

ADT requires less computational time compared to conventional dynamic-mesh CFD of a real propeller and simplifies wake flow prediction.

5.8.1 CFD Case Details

A cylindrical fluid domain of diameter 2 m and length 3 m was created with an actuator disk of diameter 0.2 m and thickness 3 mm placed 1 m from one end. Shared topology was applied so that the disk region and outer domain share the same fluid volume. This ensured that the momentum source term applied to the disk region did not create any artificial wall between the disk and the rest of the domain.

A fine mesh was generated near the disk to accurately capture the influence of the source terms, while a coarser mesh was used elsewhere. The disk was meshed using a structured method to maintain suitable values of skewness and aspect ratio.

5.8.2 Setup

- **Turbulence Model:** Spalart–Allmaras, suitable for external aerodynamic applications.
- **Boundary Conditions:** Inlet velocity of 20 m/s and atmospheric pressure at the outlet.
- **Source Terms:** These represent the propeller forces applied to the fluid. Two components were introduced:
 - **Axial source term** represents thrust and acts along the direction of flow.
 - **Swirl source term** represents rotational effects and enables the flow field to resemble that produced by an actual propeller.

To determine the most suitable source term distribution, three different axial loading approaches were tested:

- **Constant Source Term:**

$$S = \frac{T_{\text{total}}}{V_{\text{disc}}} \quad (5.23)$$

This was simple but did not match realistic thrust distribution.

- **Linear Distribution:** The source term increased radially from the centre to the edge, ensuring total thrust remained T_{total} .
- **BEM-Based Exact Distribution:** Thrust values obtained from Blade Element Momentum (BEM) analysis were applied using a UDF. This provided the most realistic representation of the propeller's radial loading.

The constant distribution was found to be an overly rough approximation, whereas the linear and BEM-based methods produced results that were much closer to expected propeller-induced flow characteristics. Based on this comparison, the BEM-derived loading was adopted for final ADT simulations, including both axial and swirl contributions.

5.8.3 Reason for Discontinuation

Although substantial effort was invested in generating the geometry, refining the mesh, implementing source terms, and examining multiple thrust distributions, the ADT approach was eventually discontinued. When integrated with the wing, ADT produced an incorrect flow field. The interaction between the actuator disk wake and the wing surface resulted in unrealistic velocity patterns and pressure distributions, which led to erroneous aerodynamic behaviour. Since this method failed to accurately represent the propeller–wing interaction, it was unsuitable for the final configuration of this study.

5.9 Virtual Blade Model

The MATLAB rotor performance prediction tool was validated using ANSYS Fluent's Virtual Blade Model (VBM), which served as the high-fidelity reference solver. The validation methodology ensured that both approaches the MATLAB Blade Element Theory (BET)-based tool and the Fluent VBM model operated under identical geometric and aerodynamic conditions. This enabled a direct assessment of model-form differences without the influence of inconsistent inputs.

5.9.1 Generation of Test Blade Geometries

A set of operating points was selected across multiple rotor radii ($R = 0.05\text{--}0.10$ m) and rotational speeds (8000–12000 RPM). For each case, the MATLAB tool generated a blade geometry specification consisting of non dimensional radial positions r/R , chord distribution $c(r)$, and geometric twist $\theta_{\text{geom}}(r)$. The collective pitch was defined as the geometric pitch at $r = 0.75R$, denoted θ_c .

For consistency with the VBM formulation, the sectional twist provided to Fluent was defined as [2]:

$$\theta_{\text{VBM}}(r) = \theta_{\text{geom}}(r) - \theta_c. \quad (5.24)$$

These geometries were used strictly as input for the validation procedure, without modifying or re-interpreting the MATLAB code.

5.9.2 Construction of the VBM Reference Model

The blade sectional data (r/R , $c(r)$, and $\theta_{\text{VBM}}(r)$) were transferred directly into the Fluent VBM Rotor Inputs interface. This ensured that the actuator-disk representation matched the MATLAB-generated geometry exactly.

Section	Radius (r/R)	Chord (m)	Twist (deg)	Airfoil File
1	0.20	0.0220	47.829	s1223_35000.dat
2	0.242	0.0213	44.932	s1223_35000.dat
3	0.284	0.0207	42.341	s1223_40000.dat
4	0.326	0.0201	37.189	s1223_40000.dat
5	0.368	0.0194	32.719	s1223_45000.dat
6	0.410	0.0188	29.016	s1223_45000.dat
7	0.452	0.0182	25.278	s1223_45000.dat
8	0.494	0.0175	20.517	s1223_50000.dat
9	0.536	0.0169	15.820	s1223_50000.dat
10	0.578	0.0163	11.820	s1223_55000.dat
11	0.621	0.0156	6.925	s1223_55000.dat
12	0.663	0.0150	3.439	s1223_55000.dat
13	0.705	0.0144	1.537	s1223_55000.dat
14	0.747	0.0137	0.000	s1223_60000.dat
15	0.789	0.0131	-1.100	s1223_60000.dat
16	0.831	0.0125	-2.400	s1223_60000.dat
17	0.873	0.0118	-3.780	s1223_60000.dat
18	0.915	0.0112	-4.890	s1223_55000.dat
19	0.957	0.0106	-5.430	s1223_55000.dat
20	1.000	0.0100	-6.100	s1223_55000.dat

TABLE 5.3: Rotor geometry entered into the VBM module.

Rotor operating parameters were matched to the MATLAB inputs, including radius, hub cutout, and rotational speed, the latter converted to angular velocity via

$$\Omega = \frac{2\pi}{60} \times \text{RPM}. \quad (5.25)$$

Prandtl's tip-loss model with tuning coefficient unity was used.

After initializing the flow field, a floating disk was generated within the VBM module, enabling Fluent to impose momentum source terms dynamically based on local velocities and the supplied airfoil polar data.

Parameter	Value
Rotor name	disk-1
VBM mode	Floating disk
Number of blades	5
Rotor speed	1500 rad/s
Rotor radius	0.075 m
Rotor root cut	0.015 m
Tip loss function	Prandtl
Tuning coefficient	1
Rotor disk origin (x, y, z)	(0, 0, 0)
Rotor disk orientation	Rotor disk normal
Disk normal vector	(-1, 0, 0)
Collective pitch	29.873°
Cyclic sine pitch	0°
Cyclic cosine pitch	0°
Cone angle	0°
Cyclic sine cone	0°
Cyclic cosine cone	0°
Rotor surface	plane-disk-disk-1, srf-diskdisk-1

TABLE 5.4: VBM rotor settings used for floating-disk simulations

5.9.3 Flow Physics, Turbulence Model, and Boundary Conditions

Air was modeled as an ideal gas, allowing density variation according to:

$$\rho = \frac{p}{RT}. \quad (5.26)$$

The k - ω SST turbulence model with default constants was initially used only for validation of the CFD setup. The inlet velocity was fixed at:

$$V_\infty = 20 \text{ m/s}, \quad (5.27)$$

with inlet turbulence quantities (computed using a turbulence intensity of 10% and a turbulence length scale of 0.001 m):

$$k = 1.609 \text{ m}^2/\text{s}^2, \quad \omega = 4412 \text{ s}^{-1}. \quad (5.28)$$

After validating the setup, the **Spalart–Allmaras (SA)** turbulence model was used for all final simulations because it provided accurate predictions for attached external aerodynamics while offering faster convergence and improved numerical stability.

For consistency, the inlet turbulence was specified using the same 10% turbulence intensity and 0.001 m length scale. The corresponding SA working variable $\tilde{\nu}$ at the inlet was computed as:

$$\tilde{\nu} = 5.3 \times 10^{-5} \text{ m}^2/\text{s}, \quad (5.29)$$

which represents the turbulent kinematic viscosity used by the model at the domain inlet.

A pressure-outlet boundary with zero gauge pressure was applied downstream. The operating pressure was set to 101 325 Pa (for takeoff) and 69 600 Pa (for cruise conditions), with the reference pressure taken at $(x, y, z) = (-0.2, 0, 0)$ to ensure numerical stability.

5.9.4 CFD Solution Procedure

The validation followed a consistent workflow. First, all flow models, turbulence settings, and boundary conditions were applied. VBM was enabled but the rotor disk was created only after initialization. Following initialization, the “Floating Disk” was generated and applied inside the VBM module, after which steady-state iterations were performed. This procedure ensured that the momentum-source distribution had reached a stable solution.

5.9.5 Extraction of Reference Thrust and Quantitative Comparison

The total thrust predicted by Fluent was obtained by integrating the axial force over the actuator disk surface. Thrust from the MATLAB tool was compared directly with this value.

The percentage deviation used for validation was defined as:

$$\varepsilon(\%) = \left| \frac{T_{\text{MATLAB}} - T_{\text{VBM}}}{T_{\text{VBM}}} \right| \times 100. \quad (5.30)$$

These deviations were plotted across rotor radii and RPM.

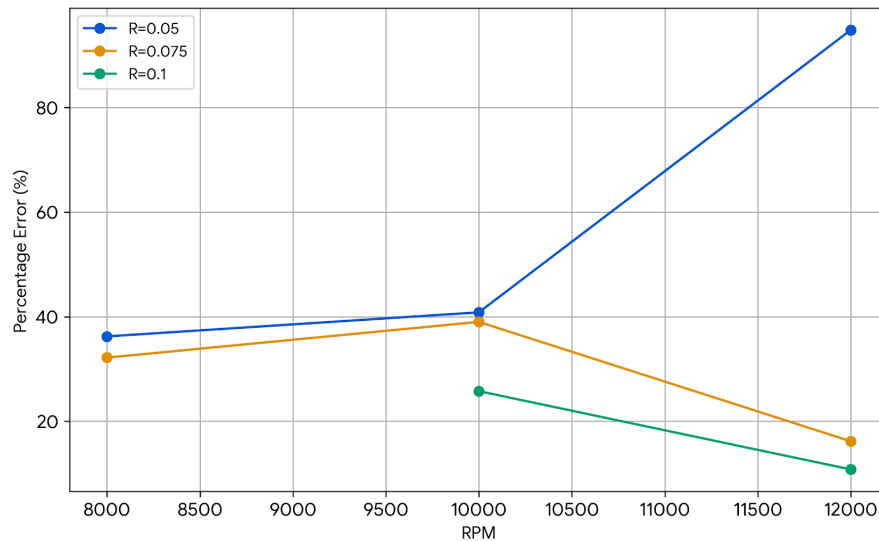


FIGURE 5.8: Percentage error between MATLAB and VBM thrust predictions.

5.9.6 Summary of the Validation Approach

The methodology ensured strict one-to-one consistency between the MATLAB generated blade geometries and the Fluent VBM representation. By enforcing identical inflow conditions, turbulence parameters, rotational speeds, and geometric inputs, the approach isolated differences arising purely from the underlying aerodynamic modeling frameworks. This provides a rigorous and repeatable basis for assessing the predictive accuracy of the MATLAB tool.

5.9.7 Results

A comprehensive set of numerical results was obtained from the VBM simulations to evaluate the aerodynamic behaviour of the rotor under the prescribed operating conditions and to facilitate a detailed comparison with the MATLAB predictions. The figures included in this section present both the integrated performance metrics (such as thrust) and the local flow characteristics around the actuator disk. These results collectively characterise the inflow, wake structure, and spanwise loading distribution predicted by the high-fidelity model.

To establish geometric and flow-field correspondence, the contours of the sectional velocity components (u , v , and w) were extracted on the plane coincident with the actuator disk. These plots allow visualisation of the orientation and symmetry of the prescribed disk loading, and provide insight into the distribution of induced velocities across the rotor radius.

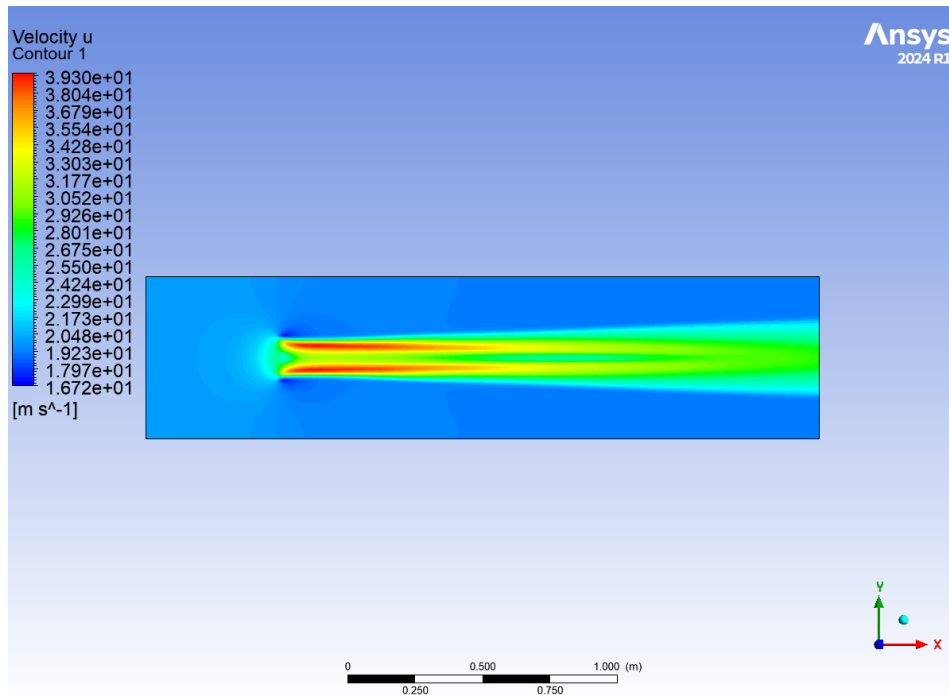


FIGURE 5.9: Contour of axial velocity component u on the rotor disk plane.

To assess the global inflow characteristics and wake contraction downstream of the rotor, contours of velocity magnitude were also extracted. These provide a direct representation of the rotor's induced flow and the distribution of momentum imparted to the fluid.

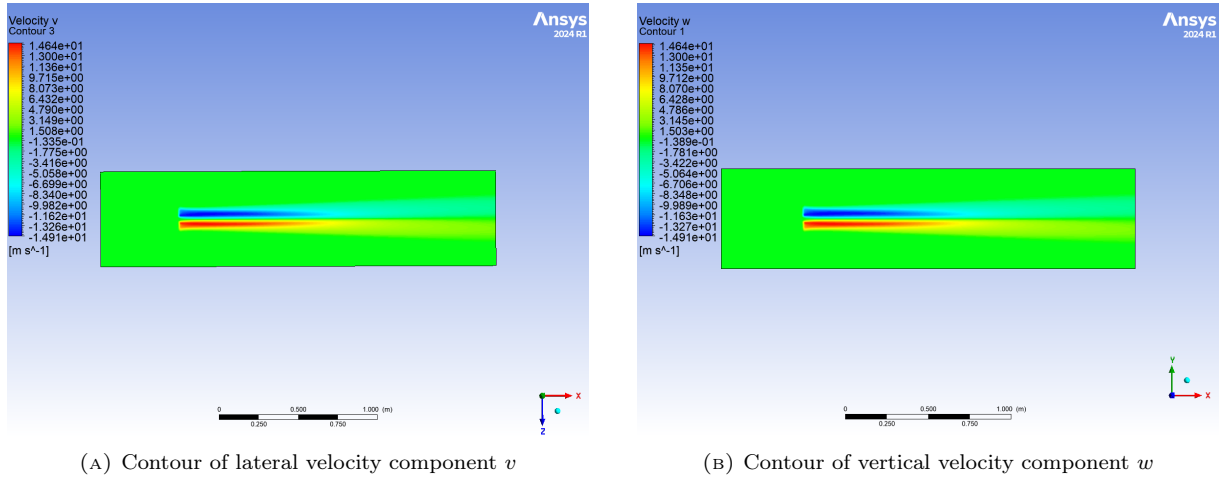


FIGURE 5.10: Velocity contours on the rotor disk plane

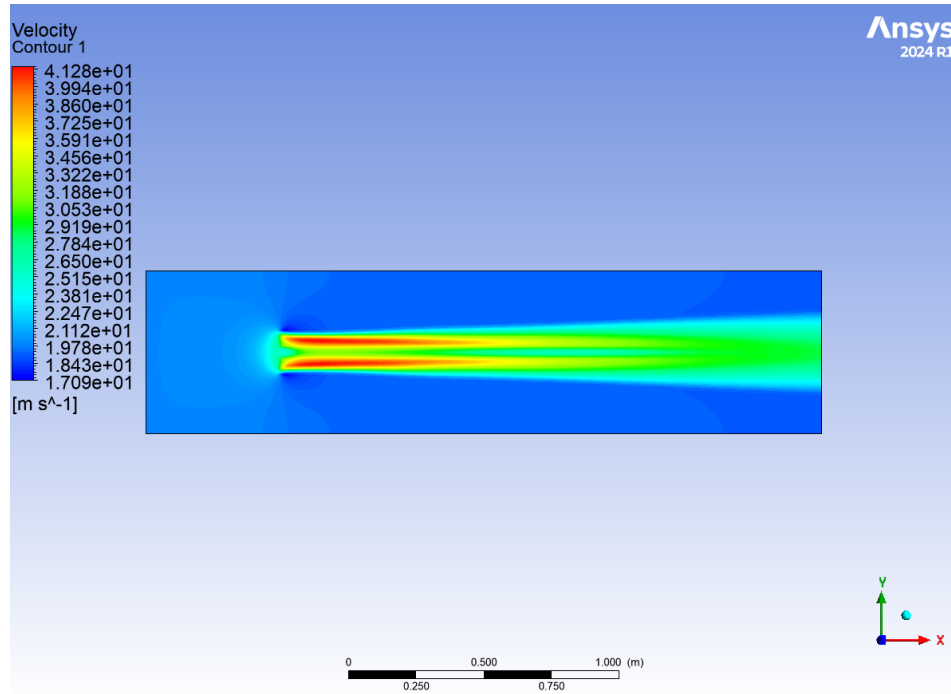


FIGURE 5.11: Velocity magnitude contours near the actuator disk.

Streamline plots were generated in the same plane to visualise the local flow deflection and streamline curvature due to rotor loading. These plots qualitatively validate the presence of accelerated axial flow through the disk and swirling motion in the wake region.

In addition to qualitative contours, quantitative outputs were collected for the purposes of validation. These included the VBM-predicted thrust values and the MATLAB-derived thrust estimates, along with the percentage deviations computed for each test case. The error distribution across rotor radius and rotational speed was plotted to evaluate the consistency of the MATLAB tool across the operating envelope.

The combination of thrust comparison, velocity-field visualisation, and streamline topology provides a unified picture of the rotor's aerodynamics under the simulated conditions. These results confirm that

the VBM model is capturing the imposed blade geometry and inflow conditions correctly, and that it serves as a reliable reference against which the MATLAB code can be validated.

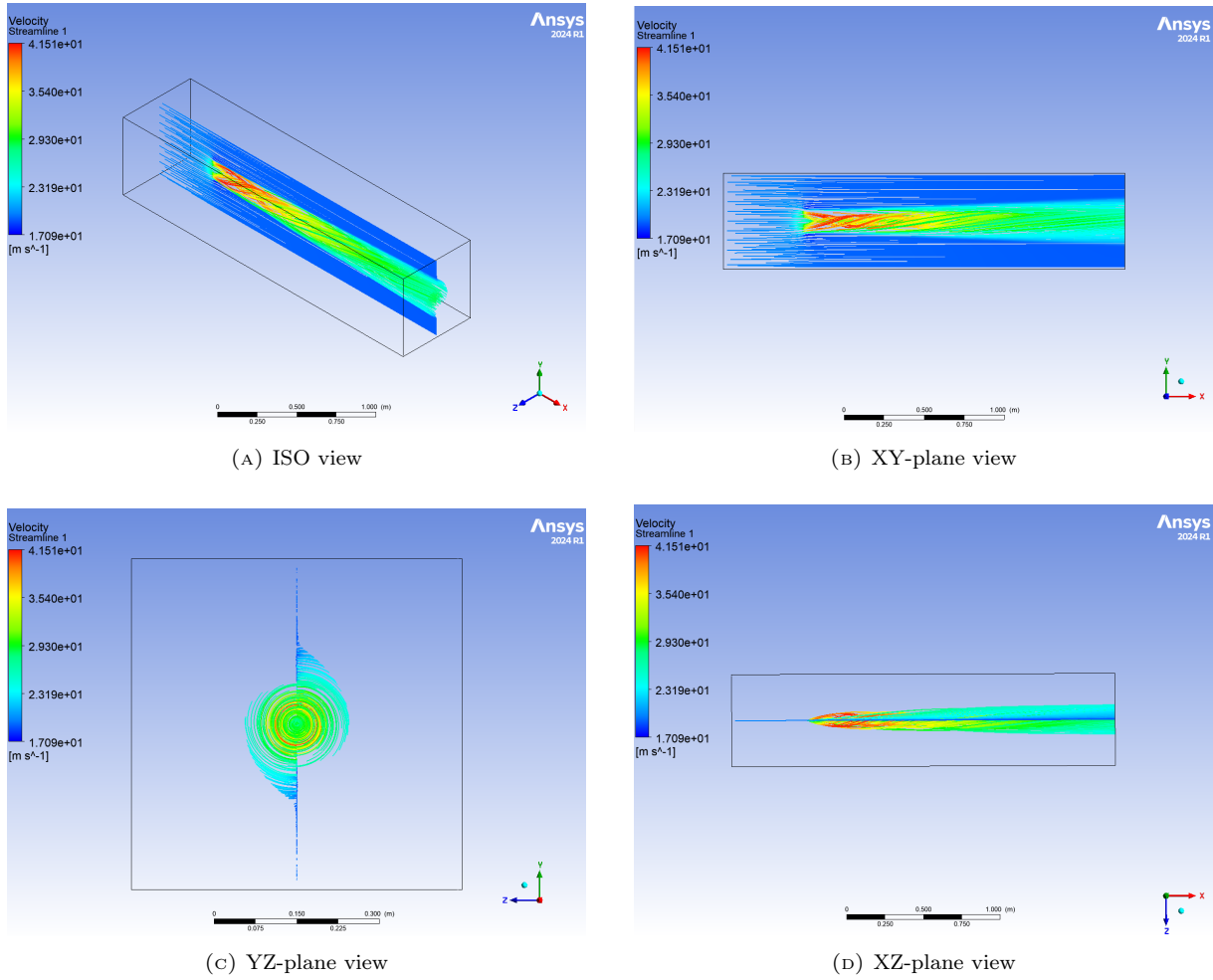


FIGURE 5.12: Streamline visualisation around the actuator disk shown from four complementary orientations: an isometric (ISO) perspective and planar slices in the XY-, YZ-, and XZ-planes.

5.9.8 Final Propeller Geometry

Looking at the possible scenarios, 4 propellers of 15cm diameter were selected, with a wing having AR of 14. The gap between each propeller was set at 3cm, and the outermost propeller centerline would be in line with the wingtip.

As per the study, the final propeller geometry has the following rotor definition:

Parameter	Value
Number of blades	5
Rotor speed (Take-off)	950 rad/s
Rotor speed (Cruise)	1200 rad/s
Rotor radius	0.075 m
Hub Radius	0.015 m
Collective pitch	35.12°

TABLE 5.5: Design Parameters of the rotor.

Section	Radius (r in m)	Chord (m)	Twist (deg)
1	0.015000	0.02550	39.336
2	0.018158	0.025079	38.947
3	0.021316	0.024658	35.878
4	0.024474	0.024237	34.149
5	0.027632	0.023816	28.726
6	0.030790	0.023395	23.388
7	0.033947	0.022974	17.749
8	0.037106	0.022553	13.978
9	0.040263	0.022132	10.347
10	0.043421	0.021711	6.5811
11	0.046579	0.021289	3.471
12	0.049737	0.020868	1.6856
13	0.052895	0.020447	0.70302
14	0.056053	0.020026	-0.000043
15	0.059210	0.019605	-0.55099
16	0.062368	0.019184	-1.3231
17	0.065526	0.018763	-4.2259
18	0.068684	0.018342	-7.0881
19	0.071842	0.017921	-8.4169
20	0.075000	0.01750	-10.575

TABLE 5.6: Rotor geometry at radial sections from the hub to the tip.

The final blade geometry obtained from the iterative design process was found to satisfactorily meet all performance targets specified in the problem statement. Using this geometry, the rotor delivers a take-off thrust of approximately **11.6 N**, achieving the required static thrust margin, while at cruise it produces about **10.2 N** of thrust, thereby meeting the prescribed cruise performance requirement. These results confirm that the designed propeller successfully satisfies the overall thrust-envelope constraints for both take-off and cruise operating conditions.

5.10 Propeller Characterization

5.10.1 Methodology and Data Acquisition

The aerodynamic performance of the designated 5-blade propeller was characterized utilizing a Virtual Blade Model (VBM) simulation framework. The simulation swept the rotor angular velocity (ω) through an operational bandwidth of 600 to 1500 rad/s to evaluate the resulting static Takeoff Thrust. Geometric inputs for the simulation included a rotor radius of 0.075 m, a root cutout of 0.015 m, and a collective blade pitch of 31.15° .

5.10.2 Thrust Response Analysis

The resulting dataset exhibits a distinct quasi-linear trend line across the tested spectrum (see Figure 5.13). While classical actuator disk theory defines thrust as proportional to the square of angular velocity ($T \propto \omega^2$), the specific interaction of the blade geometry and Reynolds numbers within this domain resulted in a highly linearized response profile. As detailed in Table 5.7, the thrust output increased monotonically from approximately 0.77 N at 600 rad/s to 15.47 N at 1500 rad/s.

Angular Velocity (rad/s)	Takeoff Thrust (N)
600	0.7726
700	1.7636
800	3.4198
900	4.8945
1000	6.2061
1100	7.7489
1200	9.3278
1300	11.2872
1400	13.3092
1500	15.4663

TABLE 5.7: Takeoff Thrust vs. Angular Velocity Data

5.10.3 Performance Implications

The absence of significant inflection points or plateaus in the thrust curve indicates that the rotor maintains fully attached flow throughout the 600–1500 rad/s envelope, effectively avoiding blade stall or critical compressibility issues at the blade tips. From a control systems perspective, this linearity is highly advantageous; it implies that the plant gain remains relatively constant, facilitating the implementation of linear control algorithms (such as PID) without the requirement for complex gain scheduling or non-linear compensation. The data confirms that the current 5-blade configuration provides a predictable and stable thrust gradient suitable for the intended propulsion requirements.

5.11 Vertical Position Study: Effect of Propeller Height Relative to Wing

5.11.1 Effect on Lift Coefficient and Pressure Distribution

When a propeller is mounted above the wing (positive vertical offset), its slipstream accelerates flow over the wing upper surface. This increased local flow velocity enhances suction on the wing's upper surface, thereby boosting pressure difference across the wing and increasing the lift coefficient C_L . Indeed,

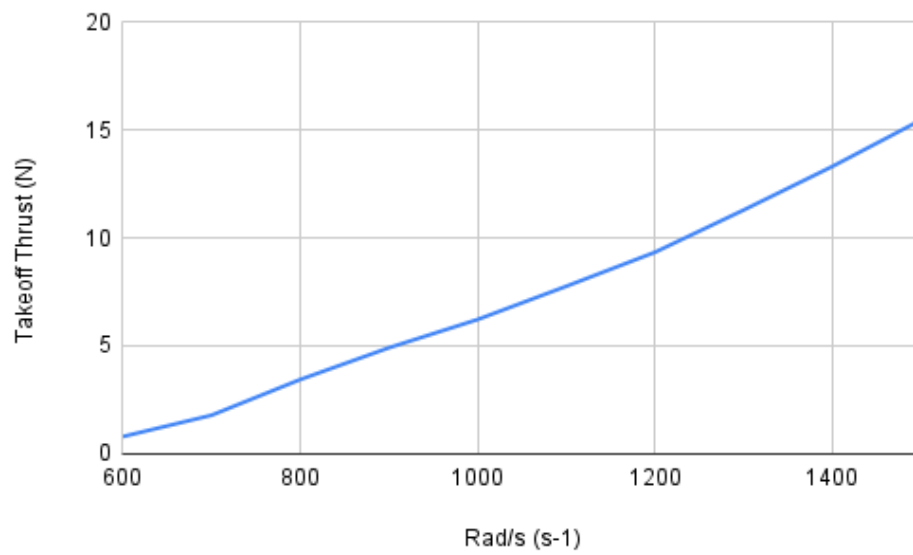


FIGURE 5.13: Plot of Takeoff Thrust (N) against Angular Velocity (rad/s), demonstrating the quasi-linear response characteristic of the propeller.

simulations of distributed-propeller configurations for UAV/DEP aircraft show a clear positive correlation between propeller height and C_L .

Conversely, when the propeller is placed below the wing (negative vertical offset), slipstream may interact adversely with the lower surface, often increasing drag and reducing the effective lift-to-drag ratio (C_L/C_D). In such cases, the benefit in lift is modest (or can even be negative for C_L/C_D), making this configuration generally sub-optimal.

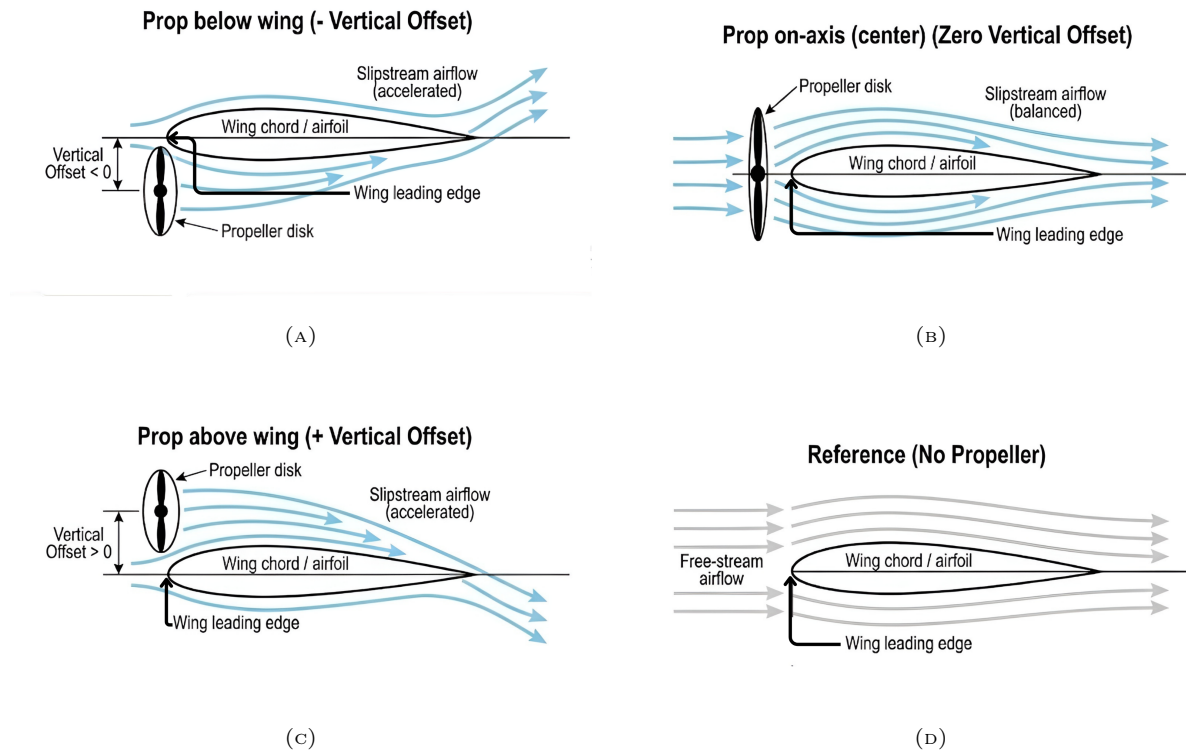


FIGURE 5.14: Vertical Placement of the propeller relative to the wing: Negative, Zero, and Positive offsets, and a case of a wing without a propeller

5.11.2 Effect on Lift-to-Drag Ratio (C_L/C_D) and Overall Efficiency

Although “above-wing” placement increases lift, the same studies indicate that drag tends to increase with vertical offset, and the wing’s overall aerodynamic efficiency (C_L/C_D) does not monotonically improve with height. In fact, for certain configurations a **center-aligned (zero vertical offset)** propeller mounting propeller disk roughly coincident with wing chord plane yields **the best** C_L/C_D . Therefore, while elevated propellers may be advantageous for high-lift conditions such as take-off (where high lift is more critical than efficiency), they may degrade cruise efficiency. A compromise is thus needed when the aircraft must perform across a wide flight envelope (take-off, climb, cruise).

5.12 Longitudinal (Chordwise) Position Study: Propeller Distance from Wing Leading Edge

5.12.1 Observations from CFD and Literature

The CFD experience indicates that when the propeller is placed very close to the wing leading edge, flow interactions between slipstream and wing leading edge create strong gradients, interference zones, and often numerical instability or non-converging solutions. This complicates design evaluation and can compromise prediction fidelity especially for multi-element or complex wing geometries.

Conversely, placing the propeller too far ahead or well separated from the wing reduces the influence of the slipstream on the wing: the accelerated flow may diffuse, diminishing beneficial lift enhancement or

flow re-energization of the wing.

Support for this behavior also appears in literature. In simulations of UAV wings using the actuator-disk method, the beneficial effect on wing lift and C_L/C_D diminishes when the axial distance between propeller and wing increases excessively.

5.12.2 Recommended Chordwise Placement

Based on both the CFD trials and literature guidance, a moderate chordwise offset approximately 10 cm aft of the wing leading edge (or as per relevant non-dimensional reference for the geometry) is recommended. This distance appears to strike a practical balance:

- It is sufficiently far to avoid numerical convergence issues or strong flow interference in CFD,
- Yet close enough for the slipstream to meaningfully interact with the wing, preserving lift augmentation and slipstream benefits.

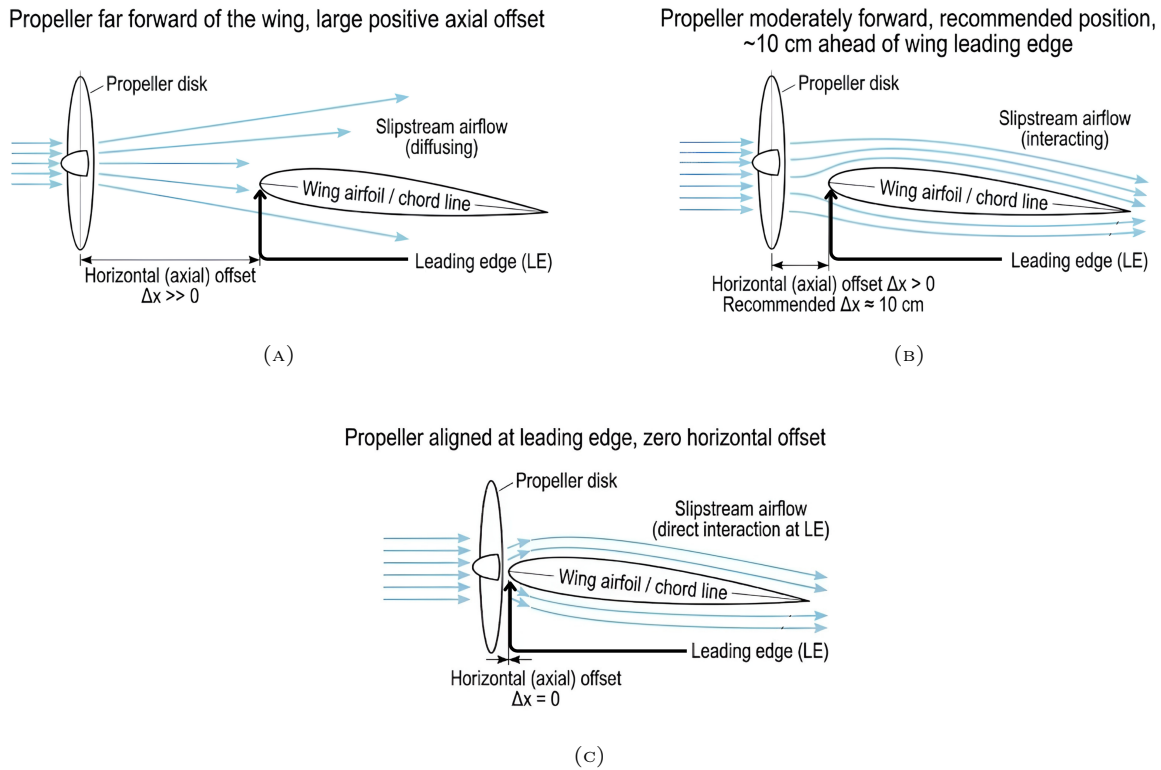


FIGURE 5.15: Longitudinal Placement of the propeller relative to the wing: Large, Small, and near-zero offsets

5.12.3 Summary

The VBM with periodic boundary conditions provided an efficient and accurate method to evaluate the complex aerodynamic coupling between the wing and distributed propeller system. By progressively refining the propeller position and flap angles, both cruise and takeoff performance were improved. The final design benefited from a synergistic combination of upper surface blowing, externally blown flaps, and controlled ducted flow, leading to enhanced lift characteristics with reduced drag penalties.

Propulsive Lift

With the rise of compact, efficient propulsion systems and the continued demand for STOL-capable aircraft, extensive work has been carried out to exploit propulsive lift to improve low-speed aerodynamic performance. Much of this development was driven by NASA in the late 20th century, focusing on delaying boundary-layer separation, improving flow attachment, and enhancing lift through active use of propeller or jet-induced flow over the wing.

6.1 Methods for Achieving Propulsive Lift

Propulsive lift techniques broadly rely on adding momentum to the airflow over the wing or modifying the local pressure distribution to maintain attached flow at high angles of attack. One widely studied approach is the **Upper Surface Blown (USB)** configuration, where a high-velocity jet is directed over the upper surface of the wing or flaps, exploiting the Coandă effect to energize the boundary layer and prevent premature separation. A related concept is the **Externally Blown Flap (EBF)**, in which the exhaust jet directly impinges on a deflected trailing-edge flap, increasing lift through jet deflection. Both USB and EBF were demonstrated effectively in aircraft such as the YC-14, YC-15, and An-72.

Another method involves **internally ducted or internally blown flaps**, where high-energy air is routed through internal passages within the wing and ejected over the upper airfoil surface. Finally, **supercirculation** which relies on generating controlled vortices or using slot-blowing to aid the overall circulation around the wing, enabling very high lift coefficients with relatively small blowing mass flow. These methods are often used in combination with multi-element wings for further enhancement of low-speed performance.

6.2 Internally Blown Flaps

The maximum lift coefficient of an airfoil is limited by stall, which occurs when the boundary layer can no longer overcome the adverse pressure gradient on the suction surface. To delay this separation and increase $C_{L_{\max}}$, internally blown flaps can be used to inject high-momentum air into the boundary layer. Instead of modeling a complete internal duct which would significantly increase meshing difficulty and computational cost, at first a simplified notch was used to represent the duct exit. This notch allowed high-velocity air to be modeled at the appropriate location on the upper surface, re-energising the flow and improving attachment under higher adverse pressure gradients, thereby increasing the achievable lift before stall.

6.3 Aerodynamic Analysis of Internally Ducted Configurations

This study evaluates the aerodynamic performance of a three-element airfoil featuring an internal duct architecture. Four distinct configurations were simulated to isolate the effects of longitudinal duct placement and third-element (flap) deflection angles on boundary layer attachment and lift-to-drag ratios (L/D).

6.3.1 Configuration Definition

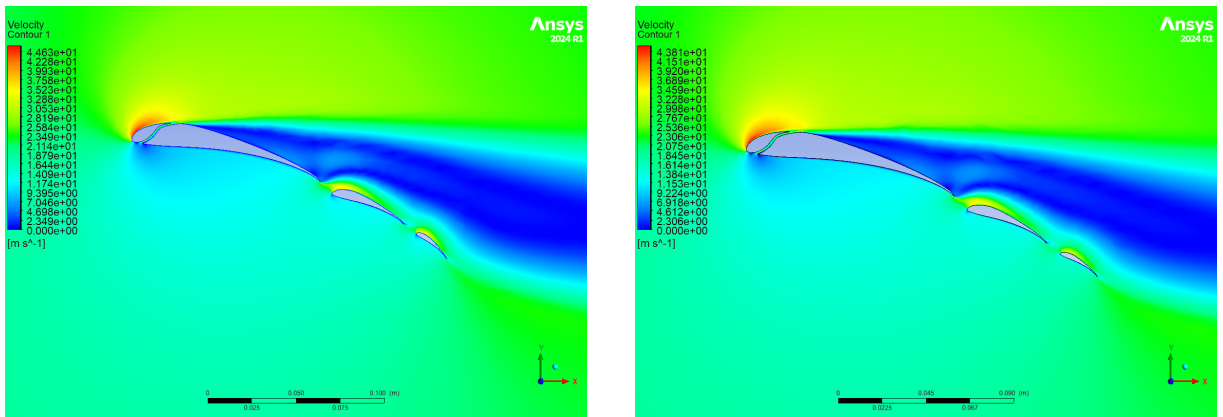
The design space was explored through two primary geometric modifications (forward vs. aft duct placement) and two operational flap settings. The test cases are defined as follows:

- **Case I (Baseline):** Forward-located internal duct with nominal third element angle.
- **Case II:** Forward-located internal duct with reduced third element deflection.
- **Case III:** Aft-located internal duct with nominal third element angle.
- **Case IV:** Aft-located internal duct with reduced third element deflection.

6.3.2 Flow Topology and Separation (Forward Duct Layout)

Figures 6.1 and 6.4 present the velocity magnitude and static pressure contours for the forward-duct configurations (Cases I and II). In Case I, the duct inlet is situated in a region of high flow acceleration near the leading edge. The contours reveal a severe adverse pressure gradient immediately downstream of the duct exit, triggering massive flow separation over the main element. This is evidenced by the large low-momentum wake region visible in the velocity streamline plot.

In Case II, the third-element flap angle was adjusted to alleviate the pressure recovery demand. While this modification resulted in a marginal recovery of the aerodynamic efficiency ($E = C_L/C_D$), the velocity contours indicate that the macro-separation induced by the upstream duct position remains the dominant flow feature, limiting the effectiveness of the flap adjustment.



(A) Case I: Velocity Magnitude (B) Case II: Velocity Magnitude

FIGURE 6.1: Velocity contours for Forward Duct configurations showing massive flow separation.

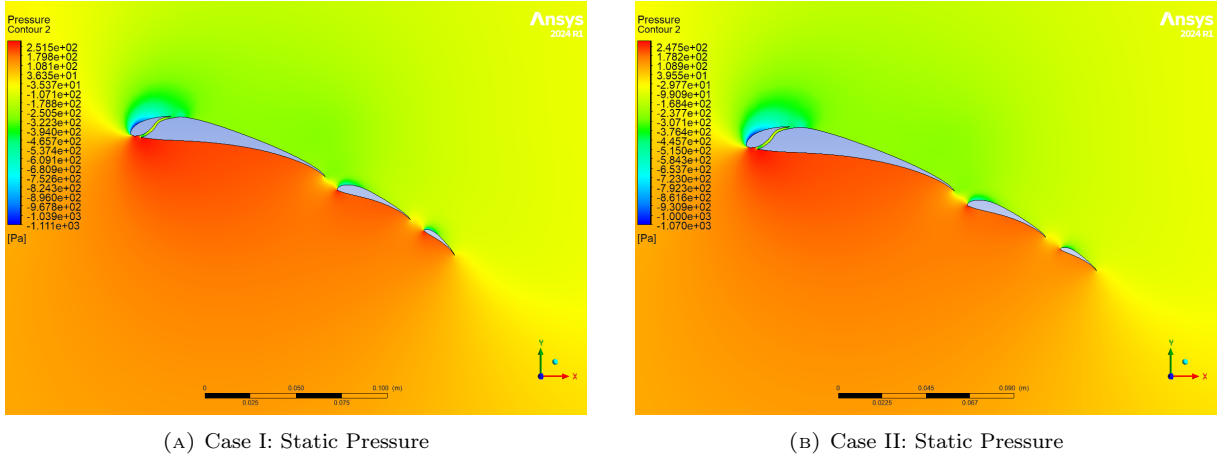


FIGURE 6.2: Static Pressure distribution for Forward Duct configurations.

6.3.3 Geometric Optimization and Attachment (Aft Duct Layout)

Based on the separation observed in the baseline, the internal duct was relocated to a significantly more aft chordwise position for Cases III and IV. Figures 6.3 and 6.4 illustrate the resulting flow field.

The aft-displacement of the duct fundamentally alters the pressure distribution. By delaying the introduction of the duct flow, the laminar-to-turbulent transition is managed more effectively, and the boundary layer remains attached over a larger percentage of the airfoil chord. The velocity contours for Case III show a substantial reduction in the wake size compared to Case I. Furthermore, the pressure contours demonstrate a more sustained suction peak, directly correlating to the observed increase in Lift Coefficient (C_L).

Case IV represents the culmination of this optimization, combining the superior aft-duct geometry with the flap deflection strategy. This configuration minimizes the pressure drag component (C_D) while maximizing circulation, resulting in the peak aerodynamic performance for this study.

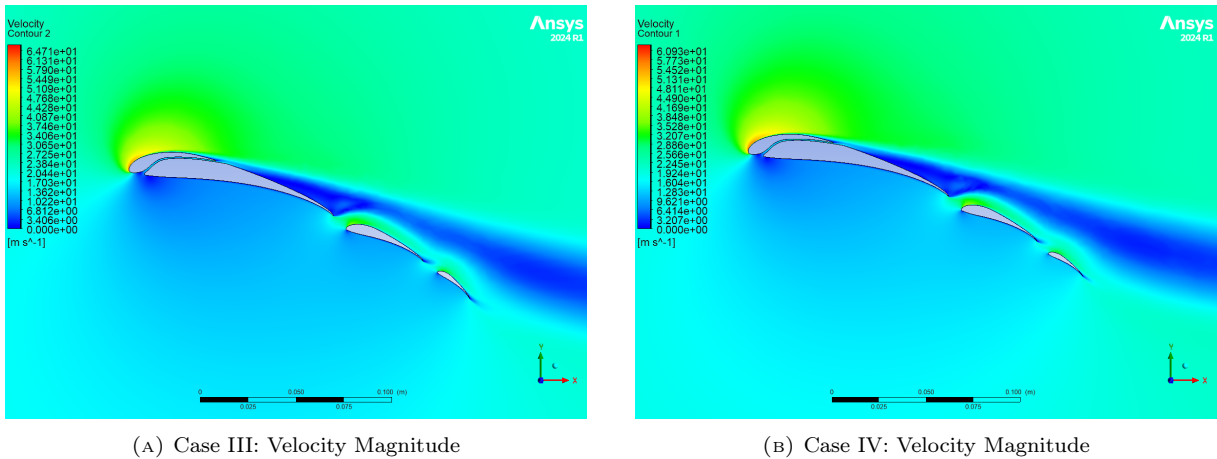


FIGURE 6.3: Velocity contours for Aft Duct configurations showing improved flow attachment.

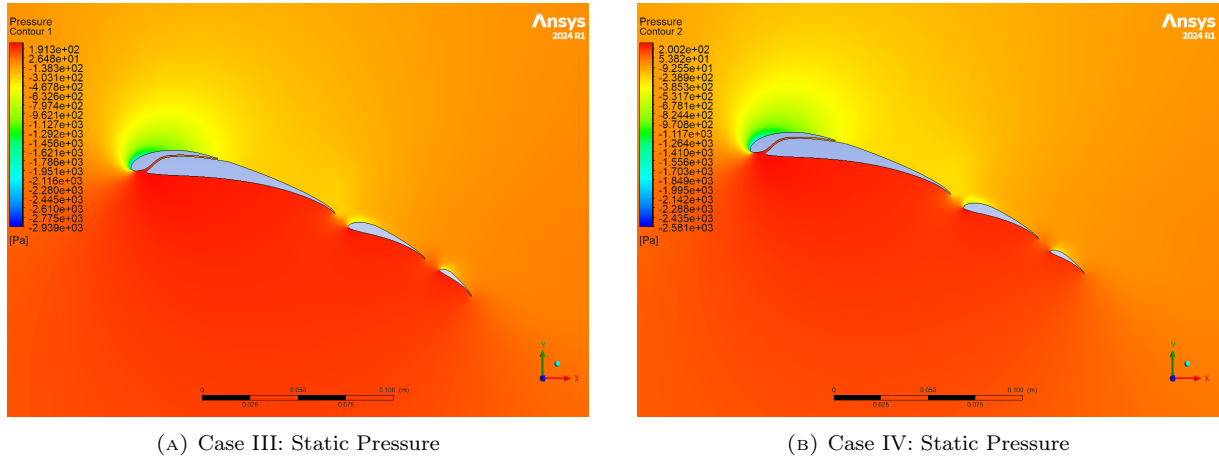


FIGURE 6.4: Static Pressure distribution for Forward Duct configurations.

6.3.4 Performance Summary

The quantitative results of the four simulated cases are summarized in Table 6.1. The transition from the forward duct (Case I) to the aft duct (Case III) yields a non-linear improvement in performance, validating the hypothesis that duct placement is the primary driver of flow stability in this multi-element airfoil.

Case ID	Duct Position	Flap Setting	C_L	C_D	C_L/C_D
Case I	Forward	Nominal (δ_1)	3.7355	0.6598	5.6616
Case II	Forward	Adjusted (δ_2)	3.5328	0.5783	6.1568
Case III	Aft	Nominal (δ_1)	4.7718	0.3050	15.6452
Case IV	Aft	Adjusted (δ_2)	4.5096	0.2923	15.4280

TABLE 6.1: Aerodynamic coefficients for ducted airfoil configurations

6.4 Concepts Explored but Not Selected

6.4.1 Trailing-Edge Duct Concept

The study investigates an innovative trailing-edge duct concept developed by the team, intended to modify circulation by redirecting a portion of the lower-surface airflow toward the upper surface. The concept draws inspiration from supercirculation principles, wherein lift enhancement can be achieved either by accelerating flow over the upper surface or by reducing flow beneath it. The proposed approach focuses on the latter by extracting lower-surface air and channeling it through an internal duct to the trailing edge.

However, this method was not developed further because the preliminary CFD results did not match our expected C_L gains.

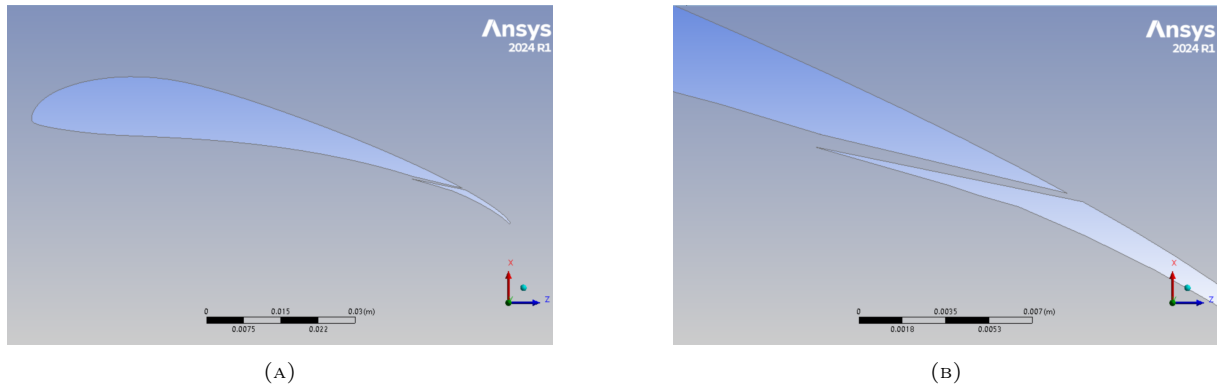


FIGURE 6.5: Geometry of Trailing-Duct

6.4.2 Vortex Generators

Vortex generators were also tested as a method of energising the boundary layer by introducing vortices near the leading edge. Although effective in medium to high Reynolds number wings, their application is limited at the low Reynolds number of our configuration (around $Re = 10^5$). At such low Re , the airfoil already exhibits a low lift-to-drag ratio (below 60), and the additional drag penalty introduced by vortex generators dominates any benefit from delayed separation. Due to this reason, they were found unsuitable for our design objectives and were not used in the final configuration.

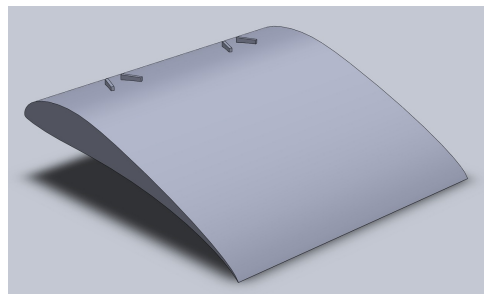


FIGURE 6.6: One by Tenth part of the wing with Vortex Generator

6.5 Wing–Propeller Aerodynamic Coupling

To accurately capture the slipstream effects of the distributed propulsion system, a Virtual Blade Model (VBM) based CFD methodology was adopted. The aircraft employs four propellers on each wing; therefore, instead of modelling the complete span, a representative wing slice containing a single propeller and its corresponding portion of the wing was simulated. This approach, combined with periodic boundary conditions, significantly reduced the overall computational cost while retaining full fidelity of the local wing–propeller aerodynamic interactions.

6.5.1 Simulation Strategy and Periodic Configuration

The distributed propulsion layout allows for a periodic description because each wing contains an equal spacing of four propeller–wing modules. By extracting one such module and applying periodic boundaries

in the spanwise direction, the flow field of an infinite repeating system can be reproduced. This reduces mesh size, simulation time, and memory requirements without compromising accuracy.

The VBM was coupled to the RANS solver through momentum and swirl source terms distributed across a thin cylindrical disc representing the propeller. This method captures the induced velocities, swirl, pressure rise, and slipstream contraction without requiring the propeller blades to be explicitly meshed.

6.6 Aerodynamic Design Optimization and Performance Sensitivity Analysis

6.6.1 Baseline Three-Dimensional Configuration and Initial Assessment

Following the successful optimization of the two-dimensional profile, the investigative focus shifted to three-dimensional (3D) computational fluid dynamics (CFD) analysis. The initial 3D geometry was generated through the spanwise extrusion of the selected 2D cross-section. In this baseline configuration, the integration of the propulsive unit was established based on preliminary theoretical guidelines: the propeller plane was situated 0.10 m (10 cm) upstream of the wing's leading edge and was aligned vertically with the leading-edge chordal line.

Subsequent simulations evaluating this baseline configuration revealed mixed performance characteristics across the operational envelope. Under the designated cruise conditions, the configuration demonstrated moderate aerodynamic efficiency, yielding a lift-to-drag ratio (L/D) of 21. However, performance shortfalls were identified during the critical takeoff phase. While the configuration generated a lift coefficient (C_L) of 6.6 and a drag coefficient (C_D) of 1.45, the resultant aerodynamic efficiency was approximately 4.55. This performance metric failed to satisfy the predefined design requirement of a takeoff $L/D \geq 5.0$, thereby necessitating iterative design modifications.

6.6.2 Investigation of Vertical Propeller Displacement

To address the efficiency deficits identified in the baseline assessment, the primary design variable selected for the first iteration cycle was the vertical position of the propulsor relative to the wing. The propeller axis was subjected to a positive vertical translation (upward displacement) of 25 mm.

Computational results for this modified configuration indicated a significant bifurcation in performance outcomes between flight regimes. The cruise condition exhibited substantial improvement, with the L/D ratio nearly doubling to a value of 40, indicating highly favorable wing-propeller interaction at lower angles of attack. Conversely, this geometric alteration proved detrimental to high-lift performance. Simulation data indicated that the upward shift of the propeller slipstream resulted in insufficient boundary layer re-energization over the flap system at high angles of attack, causing the configuration to fail in reaching the specific minimum takeoff lift target of $C_L = 6.0$.

6.6.3 High-Lift System Refinement and Stability Analysis

In response to the inability to achieve maximum lift targets in the previous iteration, a second major design revision was implemented, focusing on the geometry of the high-lift devices and the propulsor shroud. This involved a dual modification: first, the deflection angle of the third (tertiary) flap element was reduced by 7 degrees to mitigate potential flow separation at extreme deflection angles; second, the aerodynamic profile of the propulsor duct was refined to optimize intake flow quality.

Evaluations of this refined geometry showed flight conditions remain invariant. However, takeoff performance simulations indicated a marked increase in total lift generation, achieving a C_L of 7.67 with a corresponding C_D of 1.2. Despite meeting the static lift requirements, critical analysis of the flow physics revealed that these results were trending towards a transient simulation rather than a steady state. Simulations detected significant flow field unsteadiness, characterized by pronounced, periodic vortex shedding from the trailing edge and flap gaps. This indicates that at this specific high-lift operating point, the flow is aerodynamically unstable.

6.6.4 Parametric Sensitivity Study and Design Space Constraints

Following the detection of severe aero-instabilities in the high-lift configuration, an extensive parametric study was conducted to characterize the sensitivity of the design space surrounding this geometry. Iterative simulations were performed to map the effects of propulsor placement on both lift generation and flow stability. The analysis revealed exceptionally constrained operational boundaries:

1. **Vertical Sensitivity:** Parametric sweeps indicated that negative vertical displacement (shifting the propeller downward) exacerbated flow detachment issues, leading to increased amplitude of vortex shedding and numerical solution divergence. Conversely, returning to positive vertical displacement consistently resulted in a failure to achieve the requisite target lift coefficients.
2. **Longitudinal Sensitivity:** Variations in the axial spacing between the propeller and wing proved equally critical. Increasing the longitudinal separation distance resulted in a marked degradation of the propeller's ability to augment lift via slipstream impingement (C_L reduction). Conversely, reducing the separation distance (moving the propeller closer to the leading edge) induced immediate and severe flow instability due to strong potential flow interactions.

6.7 Final Geometries and results

The final geometries as selected were as follows:

Click an object. Double-click to select an edge loop. Triple-click to select a solid.

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2024 R1

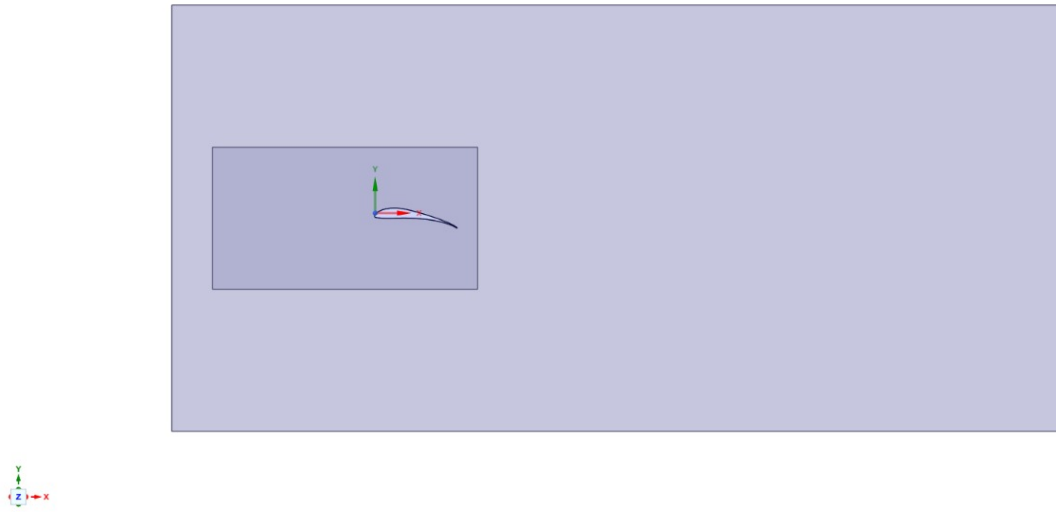


FIGURE 6.7: Final Geometry in cruise condition

Click an object. Double-click to select an edge loop. Triple-click to select a solid.

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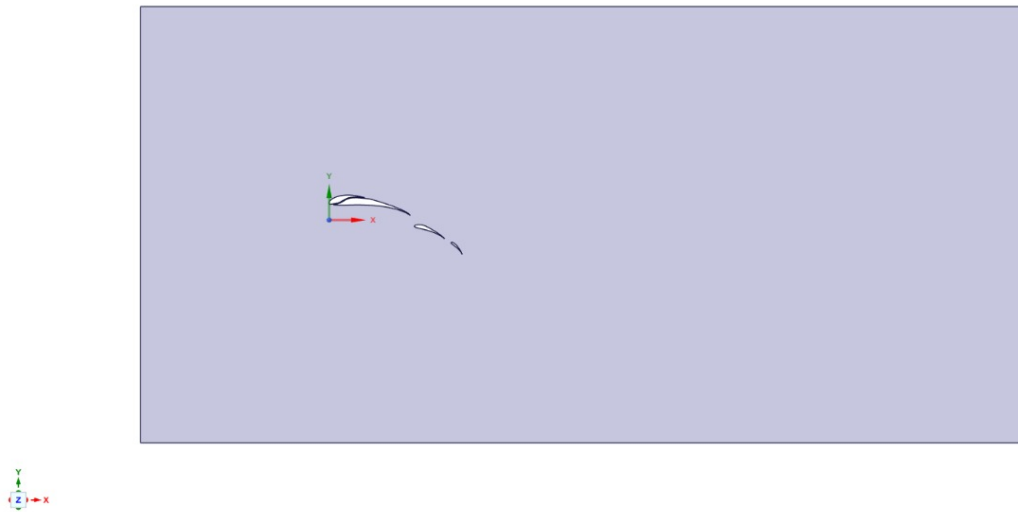


FIGURE 6.8: Final geometry in takeoff configuration

Flight Phase	C_L	C_D	C_L/C_D
Takeoff (Powered High-Lift)	6.8195	1.2695	5.377
Cruise ($\alpha = 9.25^\circ$)	1.6208	0.0802	20.209

TABLE 6.2: Aerodynamic Performance Comparison for Selected Configuration

The final location of the propeller was decided to be 0.075m ahead of the leading edge of the wing, and 0.005m vertically up from the plane of the leading edge.

Further analysis on different geometries, duct locations, propeller locations outlined the fact that flow separation and vortex shedding can be delayed by optimizing the duct location for effective re-energizing the flow. The re-energized flow now only reduces C_D while increasing C_L on the main airfoil, but energizes the flow downstream to delay separation on the third airfoil to further enhance C_L . Further analysis and transient-physics would have to be explored to further develop the flow in a correct manner. The duct effectively acts as a venturi tunnel, speeding up the incoming slipstream air from behind the propeller, to enhance flow on the upper surface of the wing.

6.8 Take-off Ground Roll Estimation

For the present propeller-driven configuration, the take-off performance was modelled using a power-based ground-roll relation explicitly formulated for propeller-driven aircraft. In such aircraft, the primary performance parameter is the power loading W/P rather than the thrust-to-weight ratio, because the propeller converts shaft power into thrust with an efficiency η_p that varies with forward speed. The selected equation assumes a constant power setting during the take-off run, a propeller operating in a low-speed, incompressible flow regime, and an approximately level, uniform runway with no slope or significant headwind. It further applies only to the ground-roll segment, i.e. from brake release until lift-off while the aircraft remains in contact with the pavement and the normal reaction on the wheels progressively decreases as lift builds up. Under these conditions, the accelerating force is governed by the balance between propulsive power, aerodynamic drag and rolling resistance, and the resulting closed-form expression provides a direct link between power loading, wing loading and required take-off distance for propeller-driven aircraft, fully consistent with the present small electric UAV configuration.

The take-off ground roll distance S_{TO} was therefore evaluated using the analytical formulation given in Sadraey's *Aircraft Design: A Systems Engineering Approach* [3], whose take-off power-loading model (Formula 4.76) relates the required power loading during take-off to aerodynamic, propulsion and ground-interaction parameters through a Bréguet-type exponential model. The corresponding expression for the required power loading is

$$\left(\frac{W}{P}\right)_{TO} = \frac{1 - \exp\left(0.6 \frac{\rho g C_{DG} S_{TO}}{W/S}\right)}{\mu - \left(\mu + \frac{C_{DG}}{C_{LR}}\right) \exp\left(0.6 \frac{\rho g C_{DG} S_{TO}}{W/S}\right)} \cdot \frac{\eta_p}{V_{TO}}, \quad (6.1)$$

where W is the aircraft weight, P the available shaft power, W/S the wing loading, ρ the air density, g gravitational acceleration, C_{DG} the drag coefficient during ground roll, C_{LR} the lift coefficient at rotation, μ the rolling friction coefficient, η_p the propeller efficiency and V_{TO} the take-off speed. The actual aircraft must satisfy

$$f(S_{TO}) = \left(\frac{W}{P}\right)_{TO} - \left(\frac{W}{P}\right)_{actual} = 0, \quad (6.2)$$

and a bisection-based numerical solver was implemented to determine the value of S_{TO} that satisfies this

equality for the given configuration. For the present design, air density was taken as $\rho = 1.225 \text{ kg/m}^3$ (ISA sea level), and $g = 9.81 \text{ m/s}^2$. The aircraft mass was set to 15 kg, giving a weight consistent with the current prototype. The wing planform area $S_{\text{wing}} = 0.10 \text{ m}^2$ yields a low wing loading representative of a small, STOL-oriented unmanned configuration. Aerodynamic coefficients during ground roll were taken as $C_{DG} = 1.0$ (a conservative/high value) and $C_{LR} = 6.5$, representing a high-lift condition at rotation. The shaft power input of 9.68 hp (converted using 745.7 W/hp) corresponds to the available power from the propulsive system, and the propeller efficiency was assumed as $\eta_p = 0.5$, which is a realistic, slightly conservative value for a small fixed-pitch propeller operating over a range of advance ratios during the ground run. The reference take-off speed was set to $V_{TO} = 20 \text{ m/s}$, consistent with the chosen combination of wing loading and lift coefficient.

Particular care was taken in selecting the rolling friction coefficient μ , as it couples tyre material, pavement condition and surface quality. Typical rolling friction coefficients for pneumatic tyres on dry, well-maintained rigid or flexible pavements (concrete/asphalt) lie in the range $\mu \approx 0.02\text{--}0.04$ for small-aircraft tyre–runway interactions [4], increasing to $\mu \approx 0.04\text{--}0.06$ for rougher asphalt or uneven surfaces, and $\mu \approx 0.06\text{--}0.10$ or higher for grass, gravel or semi-prepared strips [5]. ICAO Doc 9137—*Airport Services Manual, Part 2: Pavement Surface Conditions*—emphasises that surface texture, moisture, contamination, rubber build-up and loose material significantly influence tyre–pavement interaction and stresses the need to maintain pavement friction characteristics above prescribed thresholds to ensure safe operations on both rigid and flexible pavements [6].

For preliminary design, the choice $\mu = 0.10$ therefore intentionally biases the analysis toward a “worst credible” case, representing a small aircraft with rubber tyres operating on a marginal or semi-prepared surface rather than an ideal dry concrete runway. This ensures that the computed take-off distance is not underestimated and thus provides a conservative upper bound. With these inputs, the solver yields a take-off ground roll of approximately $S_{TO} = 29.97 \text{ m}$, representing a robust, safety-biased estimate of the required take-off distance for the current configuration.

6.9 Conclusion

The placement of the propeller relative to the wing both vertically and longitudinally has a pronounced impact on aerodynamic performance via slipstream–wing interaction. Elevated propeller mounting increases lift at the cost of drag (potentially reducing efficiency), while center-aligned vertical mounting offers a balanced lift/drag trade-off. Longitudinally, a moderate offset from the leading edge (e.g. 10 cm) is recommended to preserve slipstream benefits without incurring flow interference or numerical issues. The proposed configuration provides a practical compromise suitable for mixed flight regimes (take-off, climb, cruise).

Other Optimization Methods

High-lift optimization for STOL aircraft focuses on selecting the optimal slat and flap geometry deflection, horizontal offset, and vertical positioning to maximize lift at low speeds. This is a challenging CFD problem because the flow is dominated by strong viscous effects and complex multi-element interactions. Accurate simulation must capture several key phenomena: boundary-layer and wake mixing between elements, recirculation inside slat and flap coves, and separation on the upper surfaces at high angles of attack. Even at low freestream Mach numbers, local supersonic pockets may form near the slat leading edge. As a result, high-fidelity RANS simulations are required, with the Spalart–Allmaras model commonly used for its stability and reliability in high-lift flows.

Large geometric changes during slat/flap deployment also introduce meshing challenges. Approaches typically include full remeshing, overset (Chimera) grids, or hybrid meshes to maintain element quality throughout the motion.

For optimization, gradient-based algorithms enabled by adjoint solvers—are the most efficient, since the adjoint method provides sensitivities at a cost independent of the number of design variables. When needed, multi-point optimization combines objectives from takeoff, cruise, and landing to produce a configuration that performs well across the entire flight envelope.

7.1 Physics Informed Neural Networks

7.1.1 Background

Most commercial and open-source solvers today use the Finite Volume Method, Smoothed Particle Hydrodynamics, etc for CFD simulation. However, with the advent of Machine Learning, much work is being done on data-driven methods to speed up simulation. These methods often lack physics-based constraints and learn purely off pattern in data. A novel solution to this problem are what are referred to as PINNs or Physics Informed Neural Networks, which work on both data-driven as well as equation based constraints. This results in more physically feasible predictions rather than just pure curve-fitting. The aim here is to implement a 3D Navier-Stokes constrained PINN which is trained on a large single element airfoil CFD dataset and then extend it to multi-element airfoil systems. This would act as a mathematical model and a fast tool to help in the optimization of a high lift device based wing.

7.1.2 Theory

The dynamics of Newtonian fluids in 3D incompressible flow is governed by the Navier-Stokes equations along with the continuity equation.

$$\nabla \cdot \mathbf{u} = 0 \tag{7.1}$$

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} \right) = -\nabla p + \mu \nabla^2 \mathbf{u} + \mathbf{f} \quad (7.2)$$

where $\mathbf{u} = (u, v, w)$ is the velocity vector, p is the pressure field, and $\mathbf{f}$ represents body forces per unit volume (e.g., gravity).

The inputs to the PINN are spatial coordinates (x, y, z) from a CFD Domain and the reference data values required are the Boundary conditions to the simulation. These equations are used as physics losses or residuals which are to be optimized. The physics residuals can be written using automatic differentiation,

$$f_c = \nabla \cdot \mathbf{u}_\theta \quad (7.3)$$

$$f_x = \mathbf{u}_\theta \cdot \nabla u_\theta + \frac{1}{\rho} \frac{\partial p_\theta}{\partial x} - \nu \nabla^2 u_\theta \quad (7.4)$$

$$f_y = \mathbf{u}_\theta \cdot \nabla v_\theta + \frac{1}{\rho} \frac{\partial p_\theta}{\partial y} - \nu \nabla^2 v_\theta \quad (7.5)$$

$$f_z = \mathbf{u}_\theta \cdot \nabla w_\theta + \frac{1}{\rho} \frac{\partial p_\theta}{\partial z} - \nu \nabla^2 w_\theta \quad (7.6)$$

The total physics L_{physics} loss is given by their Mean Squared Error (MSE) over randomly sampled collocation points in the domain;

$$L_{\text{physics}} = \frac{1}{N_c} \sum_{i=1}^{N_c} (f_{c,i}^2 + f_{x,i}^2 + f_{y,i}^2 + f_{z,i}^2) \quad (7.7)$$

Along with these losses, the standard data losses for the boundary conditions. This includes the MSE for inlet velocity and outlet pressure as well as the wall residues.

$$L_{\text{data}} = \frac{1}{N_b} \sum_{i=1}^{N_b} \|\mathbf{u}_\theta(\mathbf{x}_i) - \mathbf{u}_{\text{BC}}(\mathbf{x}_i)\|^2 + \|\mathbf{p}_\theta(\mathbf{x}_i) - \mathbf{p}_{\text{BC}}(\mathbf{x}_i)\|^2 + \|\mathbf{u}_\theta(\mathbf{x}_i) - \mathbf{u}_{\text{wall}}(\mathbf{x}_i)\|^2 \quad (7.8)$$

The optimization of these residuals is done using hyperparameters, the final objective loss function is given by,

$$L_{\text{total}} = \lambda_{\text{data}} L_{\text{data}} + \lambda_{\text{physics}} L_{\text{physics}} \quad (7.9)$$

This loss function is optimized using standard gradient-descent.

7.1.3 Dataset

The model will be trained on 2-Dimensional CFD data for various airfoils, on a dataset found on Kaggle. Considering the dataset is valid only for a single element airfoil without any HLDs, the model would then be trained on the data generated by the iterative method. This training would allow efficient optimization of multiple element airfoils in an attempt to maximise the lift generated by the wing.

7.1.4 Data Processing

The training dataset used was created from 168 ANSYS Fluent simulations on 2-element wings. The (x, y) mesh data along with (u, v, p) data of each of the computational domain regions (interior, walls, inlet, outlet, and airfoil) as well as the individual flap geometry and orientation for each case were used as labels, i.e.,

$$dx, dy, \theta, d$$

where:

dx = horizontal distance between the origin and flap leading edge

dy = vertical distance between the origin and flap leading edge

θ = flap orientation angle

d = unsigned graphical distance used for region classification

Each dataset is converted into column vectors and finally into GPU tensors.

To distinguish points in the near and far fields, an unsigned distance field is constructed. All boundary points are concatenated, and for each interior/inlet/outlet/wall point, the shortest Euclidean distance to the boundary is computed. A 10% threshold is applied to define a transitional zone that captures the boundary layer.

The input data is normalized using mean-field normalization to improve training stability.

Transition-zone data are collected separately to enforce smooth blending between interior and boundary predictions.

7.2 Network Architecture

The composite model consists of two neural networks:

1. **Physics Interior Network** – learns the flow in the interior domain.
2. **Physics Boundary Network** – resolves sharp gradients and enforces boundary conditions.

The overall prediction of the composite PINN is:

$$\mathbf{U}(x, y) = \mathbf{U}_{\text{interior}}(x, y) + w(d) \mathbf{U}_{\text{boundary}}(x, y)$$

where $w(d)$ is a decaying distance-based weighting.

7.2.1 Adaptive Fourier Feature Encoding

Fluid flow PDE's lend themselves to spectral analysis, so using Fourier features enable the network to learn high-frequency flow features such as those caused due to the flaps and the sharp gradients in the

transition region

$$\phi(\mathbf{x}) = [\sin(B\mathbf{x}), \cos(B\mathbf{x})],$$

7.2.2 Geometry Encoders

The geometry data (dx, dy, theta, d) are passed through a small geometry encoder. This encoder provides for the improved resolution of local flow features and is concatenated with the Fourier features and in order to provide a full latent input to the main network.

7.2.3 Interior Domain Network

- The interior network is a deep residual MLP. The input to the network consists of multi-scale Fourier features and encoded flap geometry parameters. This is followed by a projection layer into a high dimensional latent space.
- This is followed by a sequence of Gated ResBlocks. Each block applies a two-layer MLP with batch normalization and GELU.
- The sigmoid branch learns how strongly each block's output modifies its input. Residual addition preserves gradient flow and stabilizes deep training.
- Every few blocks, additional skip connections are aggregated and added back to the main network.
- The interior network uses head outputs for the tuple (u,v,p) and these are further averaged to produce normalized predictions.

7.2.4 Boundary Zone Network

It uses a single Fourier encoder for (x,y) coordinates at a relatively high frequency to resolve sharp gradients. Uses a geometry encoder on (dx, dy, theta, d) with Tanh activations. The concatenated encodings are put through a moderate-depth fully connected network with batch normalization and Tanh. The boundary network produces a correction vector that is modulated only near the boundary with the help of an exponentially decaying weight with graphical distance.

$$w(d) = e^{-kd}$$

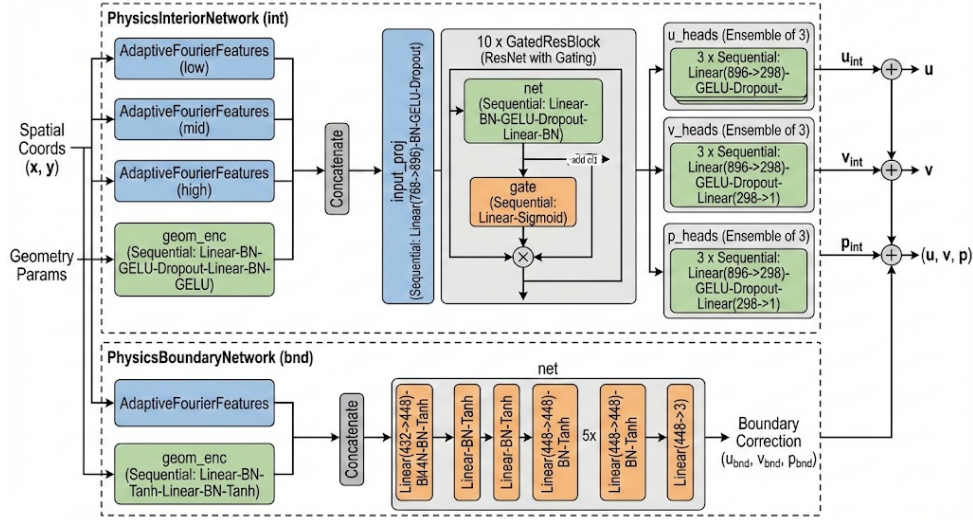


FIGURE 7.1: Consolidated PINN Structure

7.3 Loss Functions

The model is trained using a combination of data-driven and physics losses.

7.3.1 Interior Data Loss

The interior model predictions are compared against the normalized CFD data using a smooth L1 (Huber) loss,

$$\mathcal{L}_{\text{int}} = \text{Huber}(\mathbf{U}_{\text{pred}}, \mathbf{U}_{\text{CFD}})$$

This is more robust to outliers than a pure MSE and improves stability.

7.3.2 Boundary Data Loss

The data losses with respect to the boundary conditions were performed at the walls, airfoil, inlet and outlet, where

- **Inlet:**

$$\mathcal{L}_0 = \|\hat{u} - u_{\text{in}}\|^2 + \|\hat{v} - v_{\text{in}}\|^2$$

- **Outlet:**

$$\mathcal{L}_1 = \|\hat{p} - p_{\text{out}}\|^2$$

- **airfoil and wall:**

$$\mathcal{L}_{2,3} = \|\hat{u}\|^2 + \|\hat{v}\|^2$$

These are combined into a single boundary loss:

$$\mathcal{L}_{\text{bnd}} = \mathcal{L}_0 + \mathcal{L}_1 + \mathcal{L}_{2,3},$$

scaled by a factor (here 2.5) to strengthen BC enforcement.

7.3.3 Transition Data Loss

The partitioned transition zone is also subject to data losses such that the results closely match CFD data, in order to obtain smooth transition between interior and boundary predictions.

$$\mathcal{L}_{\text{trans}} = \text{Huber}_{\beta}(\hat{\mathbf{q}}(\tilde{\mathbf{x}}_{\text{trans}}) - \tilde{\mathbf{q}}_{\text{true, trans}}).$$

7.3.4 Interior Physics Loss

The final physics constraints applied to the model predictions in the interior computational domain. The losses are computed purely using the continuity equation, the continuity residual is given by,

$$r_c = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}$$

Physics loss:

$$\mathcal{L}_{\text{physics}} = E\|r_c\|^2$$

7.4 Training Strategy

The interior and boundary networks are optimized with separate Adam optimizers allowing independent learning rates. Both use cosine-annealing-with-restarts schedulers to gradually reduce the learning rate and periodically rewarm it.

To avoid destabilizing training early, the network is initially warmed up using only data losses for the first 250 epochs. After warmup, the physics losses are exponentially ramped in order to enforce continuity constraints.

For the first 250 epochs:

$$\mathcal{L} = \mathcal{L}_{\text{data}}$$

After warmup, physics loss ramps exponentially:

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda(t) \mathcal{L}_{\text{physics}}$$

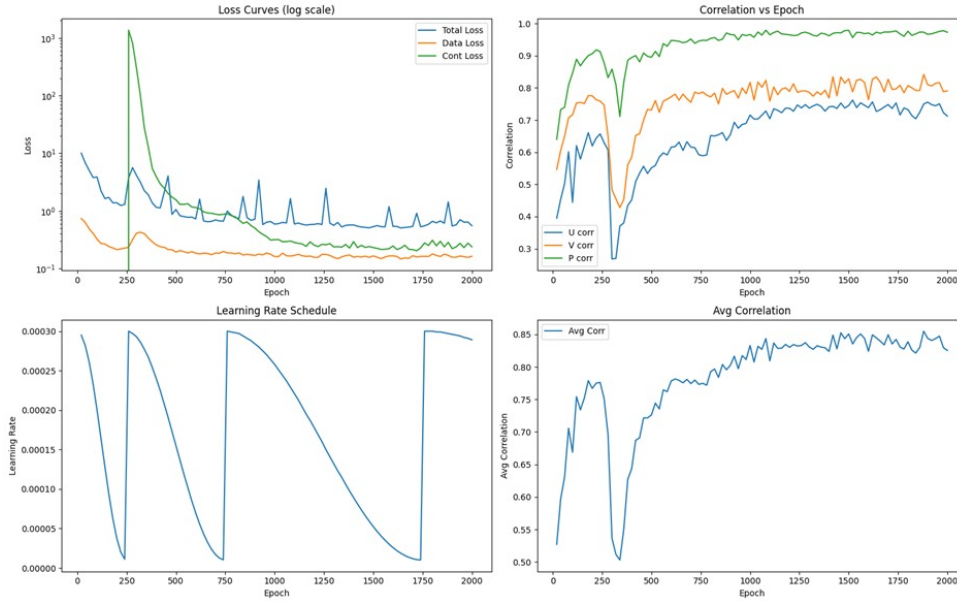


FIGURE 7.2: Training curves illustrating model loss reduction and correlation improvement over epochs.

7.5 Results and Discussion

The composite PINN was trained for a total of 1500 epochs using a ramped physics-residual strategy after the initial warm-up phase, as discussed previously. The initial PINN skeleton involved the enforcement of the full Navier–Stokes residuals; however, the highly oscillatory nature of second-order derivatives significantly hindered the training process and led to numerical instability. Consequently, the final network formulation employs only the continuity-based physics loss.

The initial unpartitioned network architecture yielded an average correlation of only 17%. However, the decomposition of the network into separate interior and boundary components, along with the introduction of a transition region, resulted in a substantial improvement in predictive accuracy. The final trained model achieved an average field-wise correlation of approximately 86%.

The trained PINN thus serves as a stand-alone surrogate mathematical model for two-element wings in two-dimensional flow. Furthermore, the framework can be generalized to multiple airfoil configurations by incorporating 8-CST geometric parameters as additional inputs, thereby enabling its use as a full-fledged parametric mathematical model.

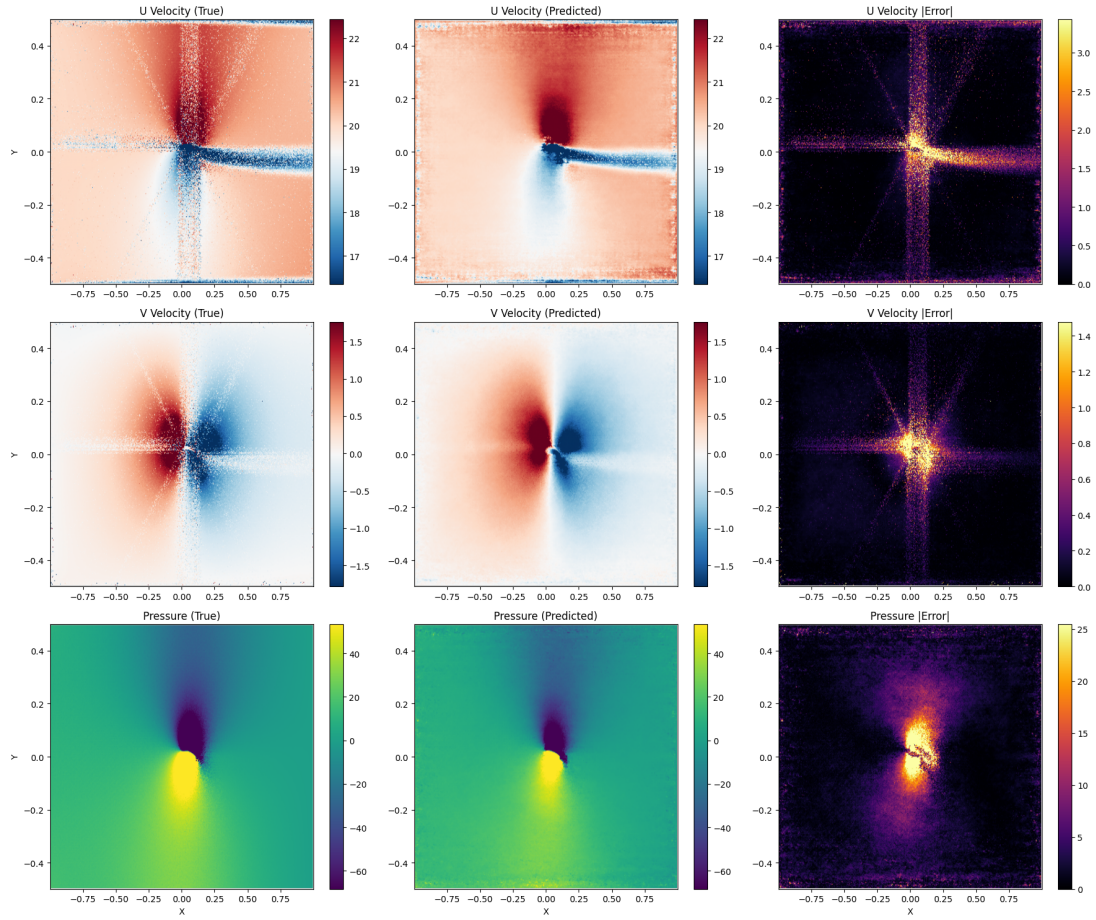


FIGURE 7.3: Field-wise comparison of streamwise velocity u , transverse velocity v , and pressure p obtained from CFD and the proposed PINN model for the two-element airfoil configuration.

7.6 Analytical Math Model

7.6.1 Jet Flap Theory

The following theory provides a foundational mathematical basis for coefficient of lift calculations over two-dimensional wings where the trailing-edge jet is deflected by an angle τ . This theory assumes inviscid, incompressible, and irrotational flow in order to treat it as a potential flow problem. Further, the jet is bounded by vortex sheets which separate it from the free stream. The internal vortices are modelled as source terms that depend on the curvature of the vortex sheet and the jet momentum flux. An integral equation is obtained for the slope of the jet, where the solution is expressed as a Fourier series to correctly account for trailing-edge behaviour. The Fourier coefficients A_0 and B_0 can be calculated as an interpolated series of $C_J^{1/2}$. This can then be used to compute the values of C_L .

$$C_L = 4\pi A_0 \tau + 2\pi(1 + 2B_0)\alpha \quad (7.10)$$

$$4\pi A_0 = 3.54 C_J^{1/2} + 0.325 C_J + 0.156 C_J^{3/2} \quad (7.11)$$

$$4\pi B_0 = 1.152 C_J^{1/2} + 1.106 C_J + 0.051 C_J^{3/2} \quad (7.12)$$

Where $C_J = J/(\rho U_\infty^2 c)$. These relations are applicable to wings with high-lift devices and propulsive lift, providing a reduced-order yet sufficiently accurate mathematical model for 2-dimensional wings of this configuration.

7.6.1.1 Theoretical Modelling

The theoretical basis for the lift produced by a jet-flapped wing is obtained by solving the flow over a two-dimensional airfoil with a thin trailing-edge jet of momentum flux per unit span (J), deflected through an angle (θ_j). The flow field is divided into two regions: the main potential flow about the airfoil, and the jet sheet (represented as a vortex sheet).

The complex potential ($w = \phi + i\psi$) satisfies Laplace's equation,

$$\nabla^2 \phi = 0.$$

Across the jet sheet, the tangential velocity experiences a discontinuity proportional to the local vortex strength (γ). Applying the Kutta condition at the trailing edge and integrating the pressure difference across the jet, Spence derived the simplified relationship:

$$C_L = 2\pi\alpha + 4\pi A_0 \tau \quad (7.13)$$

On further simplification for thin jet we obtain the previously seen expression.

A Python script was executed to extract C_J values from the `.dat` files of CFD simulations carried out in Ansys Fluent. A rough estimate of C_J was computed by capturing the trailing-edge velocity profile and mass flow rate. These values were fed into Equations 4.2 and 4.3 to determine Fourier coefficients, yielding a theoretical 2D lift coefficient ($c_{l,2d}$).

7.6.1.2 Comparison with Theoretical Predictions

Using the jet flap theory discussed previously, the lift coefficient was evaluated for the three-element duct configuration. For the operating condition corresponding to an angle of attack $\alpha = 20.55^\circ$, jet deflection angle $\tau = 9.44^\circ$, and momentum coefficient $C_J = 0.62287$, the Fourier coefficients were obtained as

$$A_0 = 0.24454, \quad B_0 = 0.12917.$$

Substituting these values into the jet flap lift relation yielded a theoretical two-dimensional lift coefficient

$$C_{L,\text{theory}} = 3.34202.$$

In comparison, the CFD simulation performed in `ANSYS Fluent` predicted a lift coefficient of

$$C_{L,\text{CFD}} = 3.90.$$

Although the theoretical estimate captures the trend of lift augmentation due to blowing, a measurable deviation exists between the two results. The following section describes the probable reasons for this discrepancy.

7.6.1.3 Possible Sources of Error

Theoretical Model Limitations The analytical formulation is based on thin-jet, inviscid, incompressible potential flow assumptions. These simplify the governing physics but neglect several important phenomena naturally captured in CFD. Specifically, the theoretical model assumes an idealized jet with no viscous loss, perfect attachment, and a fixed deflection angle, whereas the CFD solution includes jet spreading, turbulence, viscous dissipation, boundary-layer interactions, and effective three-dimensional behaviour even in nominally two-dimensional geometries. Such omissions systematically lead to an underprediction of lift when compared to high-fidelity numerical simulations.

Polynomial Fitting and Numerical Extraction of Momentum Flux The momentum coefficient C_J was obtained by fitting a seventh-degree polynomial to the CFD velocity distribution and numerically integrating the resulting profile. High-order polynomials are prone to overfitting, especially near boundaries where small fluctuations in the CFD data can induce spurious oscillations. Furthermore, integrating over the entire trailing-edge domain may inadvertently include non-jet regions,

thereby distorting the computed jet momentum flux. The absence of smoothing or filtering also allows numerical noise to propagate into the final estimate of C_J .

CFD Modelling and Numerical Effects Uncertainties associated with the CFD setup can also influence the predicted lift coefficient. The choice of turbulence model (e.g., the SA model used in this study) affects jet mixing rates and momentum transfer into the external flow, typically increasing the lift beyond inviscid theoretical values. Mesh resolution near the jet slot impacts the computed velocity gradients, while small variations in the jet exit direction within the CFD solution can lead to an effective jet angle τ that differs from the imposed geometric value. These effects contribute to a discrepancy between theoretical and numerical lift predictions.

Reference Area and Definition of Momentum Coefficient The reference area used to compute C_J in CFD may not perfectly match the assumptions of the analytical model. Even small inconsistencies in reference chord or area directly scale the computed momentum coefficient and consequently the predicted lift. Additionally, CFD inherently captures quasi-three-dimensional effects associated with finite resolution in the spanwise direction, whereas the jet flap theory assumes strictly two-dimensional behaviour.

7.6.1.4 Summary of Error Sources

A consolidated overview of the primary contributors to the discrepancy between the analytical and CFD results is provided in Table 7.1.

Category	Main Cause	Effect on C_L
Model assumptions	Inviscid, 2D potential flow assumptions	Underpredicts C_L
Polynomial fitting	Overshoot and sensitivity to noise in velocity data	Incorrect C_J estimation
Integration method	Inclusion of non-jet flow regions	Inflated or reduced C_J
Jet exit angle	Effective τ deviates from geometric value	Alters lift contribution
CFD physics	Real jet mixing, turbulence, and viscous effects	Higher C_L in CFD
Reference area	Mismatch in A_{ref} between models	Direct scaling error in C_J and C_L

TABLE 7.1: Summary of dominant error sources affecting predicted lift coefficient.

The difference between $C_{L,\text{theory}} = 3.34$ and $C_{L,\text{CFD}} = 3.90$ is consistent with expectations given the assumptions underlying jet flap theory and the numerical procedures used to extract jet momentum from CFD data. The theoretical model provides a first-order predictive tool, while the CFD solution captures the full complexity of jet-induced flow physics. Future improvements may be achieved by (i) isolating the jet core region for momentum calculations, (ii) employing lower-order polynomial or filtered fits to

reduce numerical noise, (iii) measuring the effective jet angle directly from CFD velocity vectors, and (iv) obtaining C_J from mass flow reports generated by the CFD solver for improved accuracy.

7.6.2 Veldhuis Momentum Theory for Powered-Lift Prediction

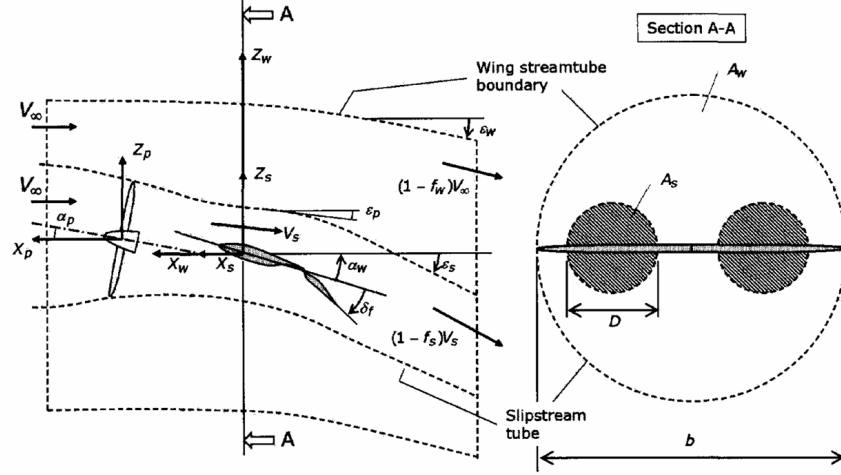


FIGURE 7.4: Veldhuis Slipstream diagram

Veldhuis' momentum theory provides a reduced-order yet physically grounded method for estimating the aerodynamic forces of wing-propeller configurations by modelling the slipstream-induced velocity increase over the lifting surfaces. The method relates the powered lift and drag coefficients to the unpowered aerodynamic characteristics and propeller slipstream parameters using momentum conservation applied to a ducted streamtube. The formulations presented here follow the procedure described by Veldhuis [7].

7.6.2.1 Governing Relations

When the propeller slipstream impinges on the wing, the local dynamic pressure increases, resulting in an augmentation of lift and a change in drag. The total powered lift coefficient is expressed as

$$C_{L,\text{powered}} = C_{Z,\text{raw}} \left(\frac{V_s}{V_\infty} \right)^2 \left(\frac{N_p S_p}{S} \right), \quad (7.14)$$

where:

- V_s is the slipstream velocity,
- N_p is the number of propellers,
- S_p is the total propeller disk area,
- S is the aircraft reference area.

The raw vertical-force coefficient is given by

$$C_{Z,\text{raw}} = (1+r)(1-f_s) \sin(\epsilon_s) + [G - (1+r)] r^2 (1-f_{w0}) \sin(\epsilon_{w0}), \quad (7.15)$$

where $r = V_\infty/V_s$ is the velocity ratio, $G = \frac{2}{N_p} \left(\frac{b}{D}\right)^2$ is the geometric slipstream–wing interaction factor, and the parameters $\epsilon_s, \epsilon_{w0}$ and f_s, f_{w0} represent slipstream deflection angles and viscous reduction factors respectively.

Determination of Wing Downwash Angle and Viscous Factor from Unpowered Aerodynamics Before evaluating the powered coefficients, Veldhuis relates the downwash angle and viscous reduction factor of the *unpowered wing* directly to its aerodynamic coefficients. Using the unpowered CFD values of C_L and C_D , the following relations provide the baseline flow-turning and viscous characteristics:

$$\epsilon_w = \arctan \left[\frac{\frac{2}{\pi A} C_L}{1 - \frac{2}{\pi A} C_D} \right], \quad (7.16)$$

$$f_w = 1 - \left(\left(\frac{2}{\pi A} C_L \right)^2 + \left(1 - \frac{2}{\pi A} C_D \right)^2 \right)^{1/2}. \quad (7.17)$$

These relations provide the wing’s effective flow deflection and viscous attenuation in the absence of slipstream effects. In the powered case, ϵ_{w0} and f_{w0} are taken from these unpowered values, ensuring consistency between baseline CFD aerodynamics and the momentum-theory augmentation model.

These expressions originate from the momentum-balance derivations on pp. 53–58 of Veldhuis’ analysis [7], where forces in the slipstream and free-stream regions are decomposed into vertical and horizontal components.

7.6.2.2 Modelling Procedure

The present implementation follows the step-by-step procedure recommended by Veldhuis:

1. **Baseline Aerodynamics:** Unpowered CFD simulations supply C_L and C_D for the clean wing.
2. **Slipstream Velocity Ratio:** The parameter $r = V_\infty/V_s$ is determined from propeller performance estimates or thrust–momentum relations.
3. **Geometric Influence Factor:** The factor G is computed from propeller diameter, wing span, and number of propellers.
4. **Slipstream Deflection and Viscous Reduction:** Normally, these parameters are extracted from Veldhuis–Chappel empirical maps for flap deflection, nacelle location, and slipstream contraction. However, these charts require highly specific geometrical inputs that are not available for the present configuration.

7.6.2.3 Zero-Order Approximation via Force Vector Summation

Because detailed empirical correlations (e.g., Chappel et al.) and static thrust test data were unavailable, a *Zero-Order Approximation* was introduced to close the model. This approximation assumes

$$\epsilon_s \approx \epsilon_{w0}, \quad f_s \approx f_{w0}.$$

This is justified by force-vector summation reasoning: in the absence of strong geometric offsets or flared ducts, the slipstream and non-slipstream portions of the wing experience similar flow turning, and viscous losses scale similarly with the local angle of attack.

Under this approximation, Eq. (7.15) simplifies to a form dependent primarily on the velocity ratio r and geometric factor G , significantly reducing parameter uncertainty while retaining physical fidelity for preliminary design.

7.6.2.4 Powered Lift and Drag Evaluation

With the simplified $C_{Z,\text{raw}}$ and the known unpowered aerodynamic coefficients, the powered lift is obtained from Eq. (7.14). An analogous expression is used to estimate powered drag:

$$C_{D,\text{powered}} = C_{D,\text{unpowered}} + C_{X,Z},$$

where $C_{X,Z}$ represents the streamwise component of the slipstream-induced force (Veldhuis Eq. 3.28)[7]

This method thus enables the computation of powered aerodynamic coefficients using only:

$$(C_L, C_D)_{\text{unpowered}}, \quad r, \quad N_p, \quad S_p, \quad G.$$

7.6.2.5 Results & Comparison

For the unpowered airfoil case, the baseline aerodynamic coefficients were obtained as:

$$C_L = 1.80, \quad C_D = 0.118.$$

Using the analytical powered-lift formulation, the predicted lift coefficient for the powered configuration was:

$$C_{L,\text{powered (analytical)}} = 2.9267.$$

The corresponding CFD simulation yielded:

$$C_{L,\text{powered (CFD)}} = 3.34.$$

The relative deviation between the analytical prediction and the CFD result is approximately:

$$\text{Error} \approx 12.4\%.$$

This indicates that while the analytical model successfully captures the lift augmentation trend, it underpredicts the lift enhancement compared to CFD, primarily due to viscous and jet-surface interaction effects not fully represented in the simplified formulation.

7.6.2.6 Sources of Deviation Between Theory and CFD

The deviation between the analytically predicted powered C_L and the CFD-observed value is well-justified and arises from a set of modelling approximations intentionally introduced to keep the analytical formulation tractable. These differences do not indicate an error in methodology; rather, they highlight the additional flow physics captured by CFD but neglected in simplified momentum-based models. The primary sources of discrepancy are:

- **Slipstream mixing and non-uniformity.** The analytical model follows the Veldhuis framework, which assumes a uniform slipstream velocity distribution. In contrast, CFD resolves the full three-dimensional structure of the propeller wake, including swirl, radial velocity gradients, and asymmetric mixing. Since these effects locally enhance circulation over the airfoil, CFD naturally predicts a higher C_L .
- **Zero-order approximations for deflection angles and loss factors.** Our calculation uses the simplifying assumptions $\epsilon_s \approx \epsilon_{w0}$ and $f_s \approx f_{w0}$ to reduce model complexity. These approximations ignore detailed flap-jet interactions and secondary flow development. CFD, however, fully resolves these effects, leading to a higher lift prediction.
- **Dependence on baseline aerodynamic coefficients.** The analytical powered-lift estimate directly scales with the unpowered C_L and C_D values. Any underprediction in the baseline CFD values—even by a small margin—propagates into the powered calculation. This makes the theoretical model conservative by design, explaining why its prediction lies below the CFD result.
- **Neglect of propeller-induced swirl in the momentum model.** The analytical formulation accounts only for axial momentum addition. CFD, on the other hand, captures the full helical slipstream, which increases effective circulation and further energizes the boundary layer. The enhanced suction peak generated by swirl contributes to the higher CFD lift coefficient.

Overall, the analytical prediction remains consistent with physical expectations: it provides a conservative estimate while CFD captures the full richness of propeller–airfoil interaction. The observed difference therefore aligns with the modelling hierarchy rather than representing a discrepancy in methodology.

7.6.2.7 Conclusion

Veldhuis’ momentum theory, combined with the Zero-Order Approximation, provides a computationally efficient method to estimate powered lift and drag using unpowered CFD results and basic propeller parameters. While less detailed than full CFD, the method accurately predicts the *trends* in powered lift augmentation and drag increase and is well suited for preliminary analysis of blown-wing configurations. The Zero-Order Approximation enables practical implementation in the absence of detailed empirical charts, allowing the theory to be applied to novel or partially characterized geometries.

Conclusion

This report documents the end-evaluation deliverables of the LAT Aerospace STOL project. The primary objective was to design a Short Take-Off and Landing (STOL) aircraft wing configuration capable of achieving high lift coefficients at low speeds without compromising cruise efficiency. The design methodology integrated theoretical modeling (Blade Element Theory, Jet Flap Theory), high-fidelity Computational Fluid Dynamics (CFD), and data-driven optimization (Physics-Informed Neural Networks).

The initial phase involved screening twenty airfoils suitable for low Reynolds number operations. The Selig S1223 was selected as the baseline profile due to its superior maximum lift coefficient ($C_{L,max} \approx 2.0$) and suction characteristics compared to the S1221. Subsequent feedback highlighted that conventional high-lift airfoils suffer significant efficiency (L/D) penalties under powered-lift cruise conditions. A secondary study compared the S1221 and S1223 at reduced angles of attack. The S1223 at $\alpha = 9.25^\circ$ was found to provide the optimal balance, maintaining higher lift generation than the S1221 while recovering the required lift-to-drag ratio for efficient cruise.

Parametric studies were conducted on trailing-edge flaps to maximize lift. A single-element flap was analyzed with chord lengths of 20%, 30%, and 40%. The 40% chord flap yielded the highest lift ($\alpha = 9.25^\circ$), while the 20% chord offered lower drag. The optimal position was identified as moderately downstream and slightly above the main trailing edge. Furthermore, to delay separation, Internally Ducted Blown Flaps (IDBF) were analyzed. Shifting the duct exit to an aft chordwise position significantly delayed boundary layer separation, achieving a $C_L = 4.77$ and an efficiency ratio L/D of **15.65**.

The 2D profiles were extruded to a 3D wing with an Aspect Ratio (AR) of 10 and a Taper Ratio (λ) of 1.0 for manufacturing simplicity. The initial baseline 3D configuration met the lift target ($C_L = 6.6$) but failed the takeoff efficiency target ($L/D = 4.55$). Subsequent refinement, involving the adjustment of the tertiary flap deflection and refining the propulsor intake, improved the takeoff C_L to **7.67** and L/D to **6.39**. However, this high-lift configuration exhibited transient vortex shedding, indicating a need for stability management in future iterations. To ensure longitudinal stability, the aerodynamic center (AC) was determined for both single and multi-element configurations. The methodology involved identifying the point where the pitching moment coefficient (C_m) remains constant with varying angle of attack. For the single element (S1223), the AC was located at approximately 23.3% chord, while for the multi-element configuration, the addition of flaps shifted the AC slightly to approximately 24.6% chord, a shift critical for sizing tail control surfaces.

A custom MATLAB script based on Blade Element Momentum Theory was developed to generate optimal propeller geometries. The design inputs included an RPM of 14,000, 4 blades, and a 0.2 m diameter. Regarding CFD implementation, Actuator Disk Theory (ADT) was initially attempted but discontinued

due to inaccurate near-field wake interactions with the wing. Consequently, a high-fidelity Virtual Blade Model (VBM) approach was adopted in ANSYS Fluent. This method accurately captured the slipstream swirl and pressure rise without the computational cost of a fully rotating mesh. Simulations confirmed that mounting propellers above the wing (positive vertical offset) enhances lift via Upper Surface Blowing (USB). A longitudinal offset of 10 cm upstream of the leading edge was recommended to prevent numerical instabilities while ensuring effective slipstream interaction.

To accelerate the optimization process, a Physics-Informed Neural Network (PINN) was developed. The architecture consists of a composite network separating "Interior" and "Boundary" flows, utilizing Fourier feature encoding to resolve sharp gradients. The model was trained on 168 CFD datasets of 2-element airfoils. In terms of performance, the model achieved an average field-wise correlation of 86% with CFD data. This surrogate model allows for rapid prediction of flow fields (velocity and pressure) for new flap configurations without running expensive RANS simulations.

The project has successfully established a high-lift configuration capable of meeting STOL requirements. **The integration of the S1223 airfoil (optimized at $\alpha = 9.25^\circ$) , aft-located internal blowing slots, and distributed propulsion has yielded a takeoff lift coefficient of 6.8195, along with a cruise L/D of 21 in such a case.** The developed analytical code, VBM-CFD, and PINNs—provides a robust framework for the final design phase, which will focus on stabilizing the high-lift flow field and structural integration.

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